

Physiology Question Bank

Total Questions: 350

Duplicate Questions Found: 27

Metric	Value
Total Questions	350
Unique Questions	323
Duplicate Questions	27

Questions

Q1. Which is the most important determinant of extracellular volume (ECF)?

- A. Total body potassium
- B. Plasma protein concentration
- C. Total body sodium ✓**
- D. Glucose concentration

Q2. What is Homeostasis?

- A. The body's immune response to infection
- B. The process of cell division and growth
- C. The maintenance of a stable internal environment ✓**
- D. The breakdown of nutrients for energy

Q3. Which neural structure receives sensory information from stomach distension after a meal via afferent nerve signals?

- A. Motor cortex
- B. Hippocampus
- C. Nucleus of the solitary tract ✓**
- D. Cerebellum

Q4. Which best describes the structural and functional unit of life?

- A. Organ
- B. Tissue
- C. Molecule
- D. Cell ✓**

Q5. Which cellular organelle is primarily responsible for energy (ATP) production in eukaryotic cells?

- A. Ribosomes
- B. Endoplasmic Reticulum
- C. Mitochondria ✓**
- D. Golgi Apparatus

Q6. Which correctly matches the junction with its major protein?

- A. Tight junction - Actin
- B. Gap junction - Connexin ✓**
- C. Desmosome - Tubulin

D. Adherens junction - Keratin

Q7. Which is the principal extracellular cation in body fluids?

- A. Potassium (K^+)
- B. Calcium (Ca^{2+})
- C. Sodium (Na^+) ✓**
- D. Magnesium (Mg^{2+})

Q8. Which compartment contains the largest proportion of total body water?

- A. Plasma
- B. Interstitial fluid
- C. Intracellular fluid ✓**
- D. Transcellular fluid

Q9. Which statement correctly differentiates intracellular from extracellular fluid composition?

- A. Extracellular fluid has a higher concentration of potassium ions.
- B. Intracellular fluid has a higher concentration of sodium ions.
- C. Extracellular fluid has a lower concentration of chloride ions.
- D. Intracellular fluid has a higher concentration of potassium ions and proteins compared to extracellular fluid. ✓**

Q10. Which electrolyte is the active transporter?

- A. Chloride (Cl^-)
- B. Potassium (K^+)
- C. Calcium (Ca^{2+})
- D. Sodium (Na^+) ✓**

Q11. Which transport process doesn't follow 'saturation kinetics'?

- A. Facilitated diffusion
- B. Primary active transport
- C. Secondary active transport
- D. Simple diffusion ✓**

Q12. Which statement best describes the Na^+/K^+ ATPase pump?

- A. Pumps 2 Na^+ into the cell and 3 K^+ out, consuming ATP.
- B. Facilitates the passive movement of Na^+ and K^+ down their concentration gradients.
- C. Pumps 3 Na^+ out of the cell and 2 K^+ into the cell, utilizing ATP. ✓**
- D. Is a secondary active transporter that co-transport Na^+ and glucose.

Q13. During the depolarization phase of an action potential, which change occurs?

- A. Voltage-gated K⁺ channels open, allowing K⁺ efflux.
- B. Voltage-gated Na⁺ channels open, allowing Na⁺ influx. ✓**
- C. Cl⁻ channels open, causing Cl⁻ influx.
- D. Ca²⁺ channels open, leading to Ca²⁺ influx.

Q14. Which ionic movement is primarily responsible for repolarization of the neuronal membrane?

- A. Na⁺ influx into the cell.
- B. Cl⁻ influx into the cell.
- C. K⁺ efflux out of the cell. ✓**
- D. Ca²⁺ influx into the cell.

Q15. Which event is responsible for the refractory period during an action potential?

- A. Increased activity of the Na⁺/K⁺ ATPase pump.
- B. Inactivation of voltage-gated Na⁺ channels. ✓**
- C. Rapid opening of voltage-gated Ca²⁺ channels.
- D. Sustained Cl⁻ influx into the cell.

Q16. The sarcomere is a functional unit which gives skeletal and cardiac muscles their striated appearance, which was first described by Van Leeuwenhoek. A sarcomere is defined as the segment between two neighboring Z-lines. They act as an anchoring point of the actin filaments. Surrounding the Z-line is the region of the I-band. I-band is the zone of thin filaments that is not superimposed by thick filaments. Named for their properties under a polarizing microscope, an A-band contains the entire length of a single thick filament. The Anisotropic band contains both thick and thin filaments. Which decreases in length during the contraction of skeletal muscle fiber?

- A. A-band
- B. Z-line
- C. I-band and H-zone ✓**
- D. M-line

Q17. Which phase of the neuronal action potential corresponds to the absolute refractory period?

- A. Hyperpolarization.
- B. Repolarization only.
- C. Depolarization and the early part of repolarization. ✓**
- D. Resting membrane potential.

Q18. A neuron fails to repolarize after an action potential. Which is the most likely cause?

- A. Excessive opening of Na⁺ channels.
- B. Increased activity of the Na⁺/K⁺ ATPase pump.
- C. Blockage of voltage-gated K⁺ channels. ✓**
- D. Overactive Cl⁻ channels.

Q19. Where is extends Pseudo H zone in a sarcomere?

- A. Within the I-band.
- B. Spanning the entire sarcomere.
- C. The central part of the A-band, containing only thick filaments. ✓**
- D. Between the Z-line and the I-band.

Q20. At rest, myosin heads are bound with adenosine diphosphate (said to be in a "cocked" position) in relation to the thin filament that does not have Ca²⁺ bound to the troponin- tropomyosin complex. Ca²⁺ bound to the troponin- tropomyosin complex induces a conformational change in the thin filament that allows for myosin heads to cross-bridge with thin filament actin. Which one of the following triggers the detachment of the myosin head from the actin filament?

- A. Release of inorganic phosphate from the myosin head.
- B. Hydrolysis of ATP into ADP + Pi.
- C. Binding of a new ATP molecule to the myosin head. ✓**
- D. Release of Ca²⁺ from troponin.

Q21. Which proteins is present in skeletal muscles, but absent in a smooth muscle?

- A. Actin
- B. Myosin
- C. Tropomyosin
- D. Troponin ✓**

Q22. Which best explains the "power stroke" during cross-bridge cycling in skeletal muscle?

- A. ATP binding to myosin, causing detachment from actin.
- B. Hydrolysis of ATP by myosin ATPase, cocking the myosin head.
- C. Release of inorganic phosphate and ADP from the myosin head, causing it to pivot. ✓**
- D. Binding of Ca²⁺ to troponin, exposing myosin-binding sites on actin.

Q23. Which statement about excitation–contraction coupling in skeletal muscle is TRUE?

- A. L-type Ca²⁺ channels on the sarcolemma directly release Ca²⁺ into the sarcoplasm.

B. Ryanodine receptors on the sarcoplasmic reticulum are activated by Ca^{2+} entering from the extracellular fluid.

C. Dihydropyridine receptors in the T-tubule membrane are mechanically linked to ryanodine receptors. ✓

D. Tropomyosin binds Ca^{2+} , initiating the conformational change that exposes myosin-binding sites.

Q24. Which event terminates skeletal muscle contraction?

A. Release of acetylcholine from the motor neuron.

B. Binding of ATP to myosin heads.

C. Active transport of Ca^{2+} back into the sarcoplasmic reticulum. ✓

D. Inactivation of acetylcholine receptors at the motor end plate.

Q25. Which protein contains the Ca^{2+} binding site that initiates cross-bridge formation in skeletal muscle?

A. Actin

B. Tropomyosin

C. Myosin

D. Troponin ✓

Q26. Which best differentiates skeletal from cardiac muscle excitation–contraction coupling?

A. Cardiac muscle requires ATP for cross-bridge cycling, while skeletal muscle does not.

B. Skeletal muscle's DHP receptors are mechanically linked to RyR channels, unlike cardiac muscle. ✓

C. Cardiac muscle utilizes troponin-tropomyosin complex, whereas skeletal muscle does not.

D. Skeletal muscle has a longer refractory period due to prolonged Ca^{2+} influx.

Q27. At the neuromuscular junction, which step directly triggers muscle fiber depolarization?

A. Acetylcholine binding to muscarinic receptors.

B. Opening of voltage-gated Ca^{2+} channels on the presynaptic terminal.

C. Activation of nicotinic acetylcholine receptors, leading to Na^{+} influx. ✓

D. Hydrolysis of acetylcholine by acetylcholinesterase.

Q28. Which type of motor neurons innervate the intrafusal fibers of striated skeletal muscle?

A. Alpha motor neurons

B. Gamma motor neurons ✓

C. Beta motor neurons

D. Sensory neurons

Q29. Which is an example of a primarily inhibitory amino acid neurotransmitter?

- A. Acetylcholine
- B. Norepinephrine
- C. GABA ✓**
- D. Serotonin

Q30. Which is relation between nerve thickness and conduction velocity?

- A. Thicker nerves have slower conduction velocity.
- B. Thinner nerves have faster conduction velocity.
- C. Nerve thickness is directly proportional to conduction velocity. ✓**
- D. Nerve thickness is inversely proportional to conduction velocity.

Q31. What is the functional unit of the nervous system?

- A. Neuron ✓**
- B. Glia
- C. Synapse
- D. Nerve fiber

Q32. Which neurotransmitters is excitatory at most synapses in the central nervous system?

- A. GABA
- B. Glycine
- C. Glutamate ✓**
- D. Acetylcholine

Q33. Which mechanisms does the hypothalamus use to maintain homeostasis?

- A. Processing sensory input for conscious perception
- B. Regulating hormone release, autonomic function, and behavioral responses ✓**
- C. Coordinating complex motor sequences
- D. Storing long-term memories

Q34. Which brain structures is primarily responsible for coordination of voluntary movement and posture?

- A. Thalamus
- B. Hippocampus
- C. Cerebellum ✓**
- D. Amygdala

Q35. Which function is primarily associated with the basal ganglia?

- A. Regulating sleep-wake cycles
- B. Initiating and modulating voluntary movement ✓**
- C. Processing emotional responses
- D. Forming new declarative memories

Q36. Which functions is primarily associated with the limbic system?

- A. Fine motor control and balance
- B. Emotion, motivation, and memory ✓**
- C. Sensory relay and conscious perception
- D. Regulation of breathing and heart rate

Q37. Across Which structures is unidirectional transport observed?

- A. Gap junction
- B. Tight junction
- C. Synapse ✓**
- D. Desmosome

Q38. Which neurotransmitter is released from preganglionic fibers of both sympathetic and parasympathetic systems?

- A. Norepinephrine
- B. Acetylcholine ✓**
- C. Epinephrine
- D. Dopamine

Q39. Which neurotransmitter is usually released by postganglionic fibers of the sympathetic nervous system?

- A. Acetylcholine
- B. Serotonin
- C. Norepinephrine ✓**
- D. Dopamine

Q40. What is the correct descending order of nerve impulse transmission velocity among the following nerve fibers?

- A. A-alpha fibers > A-delta fibers > C fibers ✓**
- B. C fibers > A-beta fibers > A-alpha fibers
- C. B fibers > A-alpha fibers > A-delta fibers
- D. A-delta fibers > B fibers > C fibers

Q41. Which is a nucleus of basal ganglia?

- A. Red nucleus
- B. Claustrum
- C. Amygdala
- D. Subthalamic nucleus ✓**

Q42. Which best explains why the effect of sympathetic stimulation is more widespread and longer lasting than that of parasympathetic stimulation?

- A. Sympathetic postganglionic neurons release acetylcholine.
- B. Parasympathetic preganglionic neurons are longer than sympathetic preganglionic neurons.
- C. Norepinephrine from the adrenal medulla is released into the bloodstream and circulates widely. ✓**
- D. Sympathetic neurotransmitters are quickly degraded by acetylcholinesterase.

Q43. Which receptors mediate relaxation of bronchial smooth muscle during sympathetic activation?

- A. Alpha-1 adrenergic receptors
- B. Beta-1 adrenergic receptors
- C. Beta-2 adrenergic receptors ✓**
- D. Muscarinic receptors

Q44. Which structure has only sympathetic innervation and lacks parasympathetic supply?

- A. Salivary glands
- B. Adrenal medulla ✓**
- C. Heart
- D. Gastrointestinal tract

Q45. Which best describes the action of the parasympathetic nervous system on the urinary bladder?

- A. Relaxes the detrusor muscle and constricts the internal urethral sphincter.
- B. Contracts the detrusor muscle and relaxes the internal urethral sphincter. ✓**
- C. Has no significant effect on bladder function.
- D. Primarily controls external urethral sphincter tone.

Q46. Which is NOT caused by sympathetic stimulation?

- A. Mydriasis
- B. Bronchodilation
- C. Increased gastrointestinal motility ✓**
- D. Glycogenolysis

Q47. A pool of presynaptic neurons innervate the dendrites of a postsynaptic neuron. Electrical signals are transferred from the dendrites to the soma of the postsynaptic neuron by which process?

- A. Action potential propagation
- B. Saltatory conduction
- C. Electrotonic conduction ✓**
- D. Synaptic transmission

Q48. Which best describes the essential components of a reflex arc in the correct sequence?

- A. Effector, efferent neuron, integration center, afferent neuron, sensory receptor
- B. Sensory receptor, afferent neuron, integration center, efferent neuron, effector ✓**
- C. Afferent neuron, sensory receptor, effector, efferent neuron, integration center
- D. Integration center, efferent neuron, sensory receptor, afferent neuron, effector

Q49. The stretch reflex (myotatic reflex) is an example of which type of reflex?

- A. Polysynaptic, contralateral
- B. Monosynaptic, ipsilateral ✓**
- C. Visceral, ipsilateral
- D. Monosynaptic, contralateral

Q50. Which sensory receptor is responsible for detecting muscle stretch and the rate of change in muscle length?

- A. Pacinian corpuscles
- B. Muscle spindles ✓**
- C. Golgi tendon organs
- D. Merkel cells

Q51. Which statements correctly describes the function of the Golgi tendon organ?

- A. Detects muscle length and rate of change in length.
- B. Initiates muscle contraction in response to stretch.
- C. Monitors muscle tension and inhibits contraction of the homonymous muscle. ✓**
- D. Facilitates muscle contraction during isometric exercises.

Q52. What is the mechanism by which calcium ions (Ca^{2+}) initiate smooth muscle contraction following their release?

- A. Binding directly to troponin C, unmasking actin binding sites.
- B. Directly phosphorylating myosin light chains, leading to cross-bridge formation.

C. Binding to calmodulin, which then activates myosin light chain kinase (MLCK) to phosphorylate myosin. ✓

D. Activating adenyl cyclase, increasing cAMP, and causing muscle contraction.

Q53. In a decerebrate animal, Which reflex changes is most characteristic?

A. Flaccid paralysis of all four limbs.

B. Enhanced flexor withdrawal reflexes.

C. Decerebrate rigidity, characterized by exaggerated extensor posture. ✓

D. Absence of all spinal reflexes.

Q54. Which fibers are primarily responsible for mediating the inverse stretch reflex?

A. Type Ia afferent fibers.

B. Type Ib afferent fibers. ✓

C. Gamma motor neurons.

D. Alpha motor neurons.

Q55. Which nerve ganglia are responsible for the secretion of saliva?

A. Ciliary and Pterygopalatine ganglia.

B. Submandibular and Otic ganglia. ✓

C. Superior cervical and Stellate ganglia.

D. Trigeminal and Geniculate ganglia.

Q56. Which is the colored part of the eye?

A. Cornea.

B. Retina.

C. Iris. ✓

D. Sclera.

Q57. Which best represents an example of a phasic receptor?

A. Nociceptor detecting constant pain.

B. Pacinian corpuscle responding to vibration. ✓

C. Muscle spindle maintaining limb position.

D. Merkel cell detecting continuous pressure.

Q58. Which sense organ is primarily responsible for detecting odors and sending signals to the brain for interpretation?

A. Retina.

B. Cochlea.

C. Olfactory epithelium. ✓

D. Gustatory papillae.

Q59. Which of these organs is used to drain fluid from the inner ear into the throat?

- A. Semicircular canals.
- B. Cochlea.
- C. Eustachian tube. ✓**
- D. Vestibule.

Q60. Which best explains how the central nervous system differentiates between modalities (touch, pain, temperature) when all sensory impulses are transmitted as action potentials?

- A. Differences in action potential amplitude.
- B. Variation in the speed of nerve impulse conduction.
- C. The "labeled line" principle, where specific pathways transmit information from distinct receptors to specific brain regions. ✓**
- D. Chemical modification of neurotransmitters based on the stimulus type.

Q61. During sensory adaptation, receptor potential amplitude gradually declines with time despite constant stimulation. Which most accurately explains this phenomenon?

- A. Decreased number of neurotransmitter vesicles in the presynaptic terminal.
- B. Inactivation of ion channels in the receptor membrane. ✓**
- C. Increased reuptake of neurotransmitters from the synaptic cleft.
- D. Depletion of intracellular ATP stores in the receptor cell.

Q62. What is the effect of increasing the intensity of a tactile stimulus in the somatosensory system?

- A. Decreased frequency of action potentials in afferent neurons.
- B. Activation of a smaller population of mechanoreceptors.
- C. Increased frequency of action potentials in afferent neurons. ✓**
- D. Shifting the perception from touch to pain.

Q63. Which receptor types is least likely to exhibit significant adaptation?

- A. Pacinian corpuscles
- B. Olfactory receptors
- C. Nociceptors ✓**
- D. Meissner's corpuscles

Q64. The potential changes that initiate action potentials in the retina are generated by the action of light on photosensitive compounds in the rods and cones. When

light is absorbed by these substances, their structure changes and this triggers a sequence of events that initiates the neural activity. Which layers of the retina does light enter the eye pass through first?

- A. Pigmented layer.
- B. Photoreceptor layer (rods and cones).
- C. Ganglion cell layer. ✓**
- D. Outer plexiform layer.

Q65. The auditory cortex, which receives impulses via the medial geniculate body, is located on which part of the brain?

- A. Occipital lobe.
- B. Parietal lobe.
- C. Frontal lobe.
- D. Temporal lobe. ✓**

Q66. The Organ of Corti on the basilar membrane extends from the apex to the base of the cochlea and thus has a spiral shape. This structure contains the highly specialized auditory receptors (hair cells) whose processes pierce the tough, membrane-like reticular lamina that is supported by the pillar cells or rods of Corti. Which ions initiate the transmission of sound by the organ of Corti?

- A. Sodium (Na^+)
- B. Calcium (Ca^{2+})
- C. Potassium (K^+) ✓**
- D. Chloride (Cl^-)

Q67. The optic nerve fibers from the nasal retina of each eye decussate at the optic chiasm. Which visual field would be lost if the right optic tract were damaged?

- A. Right nasal visual field.
- B. Left temporal visual field.
- C. Entire left visual field. ✓**
- D. Entire right visual field.

Q68. What is the primary mechanism by which the pitch (frequency) of sound is encoded in the auditory system?

- A. The amplitude of hair cell vibrations.
- B. The rate of action potentials in individual auditory nerve fibers.
- C. The specific location along the basilar membrane that is maximally stimulated. ✓**
- D. The total number of auditory nerve fibers activated.

Q69. Damage to the right auditory cortex (Heschl's gyrus) results in which deficit?

- A. Complete deafness in the right ear.
- B. Complete deafness in the left ear.
- C. Difficulty localizing sounds and understanding complex auditory patterns, especially from the left side. ✓**
- D. Impaired vision in the right visual field.

Q70. A 25-year-old student with 20/20 vision looks up from his book to view his girlfriend sitting on the other side of the room. Which is most likely to occur when the student changes his view from his book to his girlfriend?

- A. Ciliary muscles contract, and the lens becomes more convex.
- B. Ciliary muscles relax, and the lens becomes less convex (flatter). ✓**
- C. Pupil constricts to increase depth of field.
- D. Suspensory ligaments relax, increasing the curvature of the lens.

Q71. In bright light, photoreceptor adaptation (light adaptation) involves which key cellular mechanism?

- A. Increased cGMP production leading to channel opening
- B. Decreased intracellular Ca^{2+} concentration, activating guanylate cyclase
- C. Phosphorylation of photopigments and binding of arrestin ✓**
- D. Enhanced production of transducin

Q72. Lesion of Meyer's loop (temporal radiation fibers) in the visual pathway produces which specific field defect?

- A. Contralateral superior homonymous quadrantanopia ✓**
- B. Ipsilateral superior homonymous quadrantanopia
- C. Contralateral inferior homonymous quadrantanopia
- D. Bilateral temporal hemianopia

Q73. Which components primarily engages in oxygen transport in blood?

- A. Plasma proteins
- B. Hemoglobin within red blood cells ✓**
- C. Dissolved oxygen in plasma
- D. White blood cells

Q74. In Which is no counterflow mechanism observed?

- A. Renal tubules
- B. Gills of fish
- C. Lungs ✓**
- D. Testicular blood vessels

Q75. What does plasma contain that distinguishes it from serum?

- A. Albumin
- B. Immunoglobulins
- C. Fibrinogen ✓**
- D. Electrolytes

Q76. The oxygen-binding capacity of hemoglobin decreases most when Which conditions occurs?

- A. A decrease in pH
- B. An increase in 2,3-BPG
- C. Exposure to carbon monoxide ✓**
- D. A decrease in body temperature

Q77. What is the predominant type of hemoglobin present during fetal life?

- A. Hemoglobin A (HbA)
- B. Hemoglobin F (HbF) ✓**
- C. Hemoglobin A2 (HbA2)
- D. Hemoglobin S (HbS)

Q78. Which form of hemoglobin has the highest affinity for oxygen?

- A. Hemoglobin A (HbA)
- B. Hemoglobin F (HbF)
- C. Myoglobin ✓**
- D. Hemoglobin S (HbS)

Q79. Which changes occurs in RBCs of stored blood?

- A. Increased intracellular ATP and 2,3-BPG
- B. Decreased osmotic fragility and increased pH
- C. Decreased ATP and 2,3-BPG, leading to a leftward shift of the oxygen dissociation curve ✓**
- D. Increased membrane fluidity and cellular volume

Q80. Plasma is a transparent, straw-colored, liquid portion of the blood which is acquired after centrifuging blood with an anticoagulant. Serum is an undiluted fluid, the extracellular portion of blood that is acquired after centrifuging of coagulated blood. Which one of the following is present in both Serum and Plasma?

- A. Fibrinogen
- B. Clotting factors V and VIII
- C. Albumin ✓**
- D. Platelets

Q81. A woman whose blood type is A, Rh positive and a man whose blood type is B, Rh positive come to the clinic with a 3-year-old girl whose blood type is O, Rh negative. Which would be correct about the attitude of these two adults towards this child?

- A. The man is not the biological father of the child.
- B. The woman is not the biological mother of the child.
- C. Both adults are biological parents of the child. ✓**
- D. Further genetic testing is required to determine paternity and maternity.

Q82. Which hemoglobin variant shows the lowest affinity for 2,3-BPG?

- A. Hemoglobin A (HbA)
- B. Hemoglobin S (HbS)
- C. Hemoglobin F (HbF) ✓**
- D. Carboxyhemoglobin (COHb)

Q83. Which conditions would shift the oxygen-hemoglobin dissociation curve farthest to the left?

- A. Increased 2,3-BPG, acidosis, hyperthermia
- B. Decreased 2,3-BPG, alkalosis, hypothermia ✓**
- C. Increased PCO₂, acidosis, hypothermia
- D. Decreased PCO₂, normal pH, fever

Q84. Which statement correctly explains why venous blood has lower affinity for oxygen than arterial blood?

- A. Venous blood has a higher pH due to bicarbonate buffering.
- B. Venous blood has a lower temperature due to heat exchange in capillaries.
- C. Venous blood has higher PCO₂ and lower pH. ✓**
- D. Venous blood contains more fetal hemoglobin.

Q85. What primarily mediates the initial vascular response to vessel injury?

- A. Activation of the extrinsic coagulation pathway.
- B. Release of tissue plasminogen activator (t-PA).
- C. Local vasoconstriction mediated by endothelial factors and neural reflexes. ✓**
- D. Systemic hypotension caused by blood loss.

Q86. Platelets adhere to the subendothelial surface primarily through which molecular interaction?

- A. Fibrinogen binding to GP IIb/IIIa receptors.

B. Von Willebrand factor (vWF) binding to platelet GP Ib/IX/V and exposed collagen. ✓

C. Thrombin binding to PAR-1 receptors.

D. Tissue factor initiating the extrinsic coagulation pathway.

Q87. Which platelet receptors mediates platelet–platelet aggregation?

A. Glycoprotein Ib (GP Ib)

B. P2Y₁₂ receptor

C. Glycoprotein IIb/IIIa (GP IIb/IIIa) ✓

D. PAR-1 receptor

Q88. Which best explains why Vitamin K deficiency leads to a prolonged prothrombin time (PT) before the activated partial thromboplastin time (aPTT)?

A. Factor VII has a shorter half-life than Factor IX. ✓

B. The extrinsic pathway is exclusively dependent on Vitamin K-dependent factors.

C. The aPTT measures only factors independent of Vitamin K.

D. Vitamin K primarily affects the activation of thrombin in the intrinsic pathway.

Q89. What is the effect of thrombin binding to PAR-1 receptors during platelet activation?

A. Directly initiates fibrinolysis.

B. Inhibits platelet aggregation.

C. Induces intracellular signaling leading to platelet activation and aggregation. ✓

D. Promotes platelet adhesion to exposed collagen.

Q90. Which mechanisms provides negative feedback regulation in the coagulation cascade?

A. Activation of the intrinsic pathway by Factor XII.

B. Thrombin's conversion of fibrinogen to fibrin.

C. Activated Protein C inactivating Factors Va and VIIIa. ✓

D. Platelet adhesion to exposed collagen.

Q91. Which describes the appearance of erythrocytes under a microscope?

A. Large, spherical cells with a prominent nucleus.

B. Small, biconcave discs lacking a nucleus. ✓

C. Irregularly shaped, phagocytic cells with granules.

D. Spindle-shaped cells with contractile proteins.

Q92. What is the best description of ABO blood group antigens?

A. Antibodies found in the plasma that react with foreign red blood cells.

- B. Proteins located within the cytoplasm of platelets, involved in clotting.
- C. Glycoproteins or glycolipids expressed on the surface of red blood cells. ✓**
- D. Receptor molecules found on the surface of white blood cells, mediating immune responses.

Q93. Which statement best explains the universal donor property of O negative blood?

- A. It contains both anti-A and anti-B antibodies, which can neutralize recipient antigens.
- B. It has an absence of A, B, and Rh antigens on the red blood cell surface. ✓**
- C. Its red blood cells have a unique membrane structure that prevents agglutination.
- D. It produces a special type of hemoglobin that is compatible with all blood types.

Q94. In a patient with severe liver disease, Which blood group antigens is most likely to show reduced expression on RBCs?

- A. Rh antigens.
- B. ABO antigens.
- C. Kell antigens.
- D. Lewis antigens. ✓**

Q95. Why is ABO incompatibility more likely to cause severe hemolysis compared to Rh incompatibility?

- A. ABO antigens are more abundant on the red blood cell surface.
- B. ABO antibodies are primarily IgM, which are highly efficient at activating complement. ✓**
- C. Rh antibodies are typically IgA, which are less potent hemolytic agents.
- D. Rh antigens are less immunogenic than ABO antigens.

Q96. Which genetic mutations would result in the Bombay blood group phenotype?

- A. A mutation in the ABO gene causing non-functional A or B transferase.
- B. A homozygous mutation in the H gene (FUT1) leading to the absence of H antigen. ✓**
- C. Duplication of the RhD gene, increasing D antigen expression.
- D. A mutation leading to overexpression of the secretor gene (FUT2).

Q97. What are the "Gate" cells in the heart?

- A. Sinoatrial (SA) node cells, due to their role as the primary pacemaker.
- B. Atrioventricular (AV) node cells, which delay impulse transmission to the ventricles. ✓**
- C. Purkinje fibers, responsible for rapid conduction throughout the ventricles.
- D. Cardiac contractile myocytes, which generate the main pumping force.

Q98. In which part of the circulation does the greatest drop in systemic arterial blood pressure occur?

- A. Large elastic arteries (e.g., aorta).
- B. Capillaries.
- C. Arterioles. ✓**
- D. Vena cava.

Q99. Which ion plays the most critical role in maintaining the plateau phase (Phase 2) of the cardiac muscle action potential?

- A. Sodium (Na^+).
- B. Potassium (K^+).
- C. Calcium (Ca^{2+}). ✓**
- D. Chloride (Cl^-).

Q100. During which phase of the cardiac cycle are both the atrioventricular (AV) and semilunar valves closed while the ventricular volume remains constant?

- A. Ventricular ejection.
- B. Isovolumetric contraction. ✓**
- C. Ventricular filling.
- D. Atrial systole.

Q101. What is the effect of vagus nerve stimulation during the cephalic phase of digestion?

- A. Inhibition of gastric acid secretion.
- B. Increased pancreatic bicarbonate secretion.
- C. Stimulation of gastric acid and pepsinogen secretion. ✓**
- D. Relaxation of the lower esophageal sphincter.

Q102. What is the major underlying mechanism responsible for impaired gastric emptying of liquids following vagotomy?

- A. Reduced frequency of antral contractions.
- B. Enhanced pyloric sphincter opening.
- C. Loss of receptive relaxation of the fundus. ✓**
- D. Increased duodenal feedback inhibition.

Q103. Which statements regarding saliva is true under resting conditions?

- A. It is hypertonic relative to plasma.
- B. Its primary electrolyte composition is identical to plasma.
- C. It contains enzymes for both carbohydrate and lipid digestion. ✓**
- D. Its flow rate is maximally stimulated.

Q104. What does the “Law of the intestine” refer to?

- A. The regulation of nutrient absorption in the ileum.
- B. The mechanism of haustral churning in the colon.
- C. Coordinated propulsion of chyme involving contraction proximal and relaxation distal to a bolus. ✓**
- D. The role of bacterial flora in vitamin synthesis.

Q105. Which most accurately explains why saliva becomes hypotonic as it leaves the duct?

- A. Active secretion of water into the ducts.
- B. Greater reabsorption of solutes (Na⁺, Cl⁻) than secretion of solutes (K⁺, HCO₃⁻) in the ductal cells, with low water permeability. ✓**
- C. Primary saliva being initially hypotonic in the acini.
- D. High water permeability of the ductal cells allowing water to exit faster than solutes.

Q106. Which hormone primarily inhibits gastric emptying by acting on the pyloric sphincter when fatty chyme enters the duodenum?

- A. Gastrin.
- B. Secretin.
- C. Cholecystokinin (CCK). ✓**
- D. Motilin.

Q107. In absence of carbonic anhydrase activity in parietal cells, which process is most affected?

- A. The transport of chloride ions into the gastric lumen.
- B. The generation of hydrogen ions for acid secretion. ✓**
- C. The secretion of intrinsic factor.
- D. The synthesis of carbonic anhydrase itself.

Q108. When chyme containing high acid content enters the duodenum, Which responses is most accurately expected?

- A. Increased gastric acid secretion.
- B. Release of cholecystokinin (CCK).
- C. Stimulation of pancreatic bicarbonate secretion. ✓**
- D. Accelerated gastric emptying.

Q109. Which cells is the primary source of intrinsic factor, and what is its physiologic significance?

- A. Chief cells; facilitates fat absorption.

- B. G cells; stimulates gastric acid secretion.
- C. Parietal cells; essential for vitamin B12 absorption. ✓**
- D. Enterochromaffin-like (ECL) cells; promotes histamine release.

Q110. During the intestinal phase of digestion, amino acids and peptides in the duodenum mainly stimulate which hormone to enhance pancreatic enzyme secretion?

- A. Secretin.
- B. Gastrin.
- C. Cholecystokinin (CCK). ✓**
- D. Vasoactive Intestinal Peptide (VIP).

Q111. Which mechanisms explains glucose absorption from the intestinal lumen into enterocytes?

- A. Primary active transport by Na⁺/K⁺-ATPase.
- B. Facilitated diffusion via GLUT5.
- C. Secondary active transport via SGLT1 and facilitated diffusion via GLUT2. ✓**
- D. Simple diffusion directly across the lipid bilayer.

Q112. Which statements regarding gastric juice composition is correct?

- A. Parietal cells secrete pepsinogen.
- B. Chief cells secrete intrinsic factor.
- C. Hydrochloric acid is secreted by parietal cells. ✓**
- D. Gastrin is a primary component of gastric juice.

Q113. Which best describes the physiological role of enterokinase (enteropeptidase)?

- A. It cleaves peptide bonds of proteins in the stomach.
- B. It activates chymotrypsinogen to chymotrypsin.
- C. It converts trypsinogen into active trypsin. ✓**
- D. It facilitates the absorption of vitamin B12.

Q114. A patient with steatorrhea has low levels of pancreatic lipase but normal bile secretion. Which step of fat digestion or absorption is primarily impaired?

- A. Emulsification of fats.
- B. Formation of chylomicrons.
- C. Digestion of triglycerides. ✓**
- D. Transport of digested fats into the lymphatic system.

Q115. Which transporters mediate fructose absorption across the apical membrane of intestinal mucosa?

- A. SGLT1.
- B. GLUT2.
- C. GLUT5. ✓**
- D. Na⁺/K⁺-ATPase.

Q116. Which accurately explains the role of bile salts in digestion and absorption?

- A. They directly break down triglycerides into fatty acids and glycerol.
- B. They are enzymes that digest carbohydrates.
- C. They emulsify fats and form micelles to aid in fat absorption. ✓**
- D. They neutralize gastric acid in the duodenum.

Q117. Which pancreatic enzyme is secreted in an inactive form and later activated in the intestinal lumen?

- A. Amylase.
- B. Lipase.
- C. Trypsinogen. ✓**
- D. DNAase.

Q118. In the regulation of pancreatic secretion, which hormone is most responsible for stimulating bicarbonate-rich fluid secretion from pancreatic ductal cells?

- A. Gastrin.
- B. Cholecystokinin (CCK).
- C. Secretin. ✓**
- D. Vasoactive intestinal peptide (VIP).

Q119. Which function of the liver is not directly related to digestion?

- A. Production of bile.
- B. Synthesis of albumin.
- C. Metabolism of absorbed nutrients.
- D. Detoxification of drugs. ✓**

Q120. A patient has obstruction of the common bile duct. Which consequence is most likely?

- A. Impaired gastric emptying due to reduced secretin release.
- B. Increased absorption of fat-soluble vitamins.
- C. Steatorrhea and jaundice. ✓**
- D. Hyperglycemia due to decreased insulin secretion.

Q121. Which substrates yields the highest respiratory quotient (RQ) upon oxidation?

- A. Fatty acids
- B. Glucose ✓**
- C. Amino acids
- D. Ketone bodies

Q122. What process accounts for the major energy expenditure in a resting adult human body?

- A. Muscle contraction
- B. Hormone synthesis
- C. Digestion and absorption
- D. Na⁺/K⁺ ATPase activity ✓**

Q123. What doesn't lie in renal medulla?

- A. Loop of Henle
- B. Collecting duct
- C. Glomerulus ✓**
- D. Vasa recta

Q124. Which is not an effect of Angiotensin II?

- A. Vasoconstriction
- B. Increased aldosterone secretion
- C. Decreased thirst sensation ✓**
- D. Increased ADH secretion

Q125. What is the approximate glomerular filtration rate (GFR) in a healthy adult?

- A. 25 ml/min
- B. 60 ml/min
- C. 125 ml/min ✓**
- D. 250 ml/min

Q126. Where does urine acidification occur?

- A. Glomerulus
- B. Proximal convoluted tubule
- C. Loop of Henle
- D. Collecting duct ✓**

Q127. Which is responsible for neutralization hydrochloric acid?

- A. Pepsin
- B. Gastrin
- C. Secretin
- D. Bicarbonate ✓**

Q128. Which part of the nephron is impermeable to water under all physiological conditions?

- A. Proximal convoluted tubule
- B. Descending limb of Loop of Henle
- C. Thick ascending limb of Loop of Henle ✓**
- D. Collecting duct

Q129. Experimental studies are conducted to study renal tubular transport proteins. A new drug is developed that specifically and completely inhibits glucose transport in proximal renal tubules. Which substances has the same clearance as these drug?"

- A. Inulin ✓**
- B. Creatinine
- C. Para-aminohippurate (PAH)
- D. Urea

Q130. Which is effective renal plasma flow (in ml/ min) in human's?

- A. 50-100 ml/min
- B. 120-150 ml/min
- C. 200-300 ml/min
- D. 550-650 ml/min ✓**

Q131. Which changes would increase glomerular filtration rate (GFR) the most?

- A. Constriction of the afferent arteriole
- B. Dilation of the efferent arteriole
- C. Increase in Bowman's capsule hydrostatic pressure
- D. Dilation of the afferent arteriole ✓**

Q132. A decrease in systemic blood pressure is sensed by the juxtaglomerular apparatus (JGA). Which is the most direct effect of this signal?

- A. Increased sodium reabsorption in the distal tubule
- B. Increased antidiuretic hormone (ADH) secretion
- C. Renin release from juxtaglomerular cells ✓**
- D. Decreased glomerular filtration rate (GFR)

Q133. Why do plasma urea and creatinine levels increase in chronic kidney disease?

- A. Increased production of urea and creatinine
- B. Decreased reabsorption of urea and creatinine
- C. Reduced glomerular filtration rate (GFR) ✓**
- D. Increased tubular secretion of these substances

Q134. Which is true regarding renal autoregulation of GFR?

- A. It primarily involves hormonal control by ADH.
- B. It maintains a relatively constant GFR over a wide range of mean arterial pressures. ✓**
- C. It relies solely on the sympathetic nervous system.
- D. It increases GFR when systemic blood pressure falls below 80 mmHg.

Q135. What is the main function of the renin–angiotensin–aldosterone system (RAAS)?

- A. Lowering blood pressure and inhibiting sodium reabsorption.
- B. Regulating blood glucose levels.
- C. Increasing blood pressure and blood volume. ✓**
- D. Stimulating erythropoietin production.

Q136. Which will stimulate renin secretion?

- A. Increased systemic blood pressure
- B. Increased sodium delivery to the macula densa
- C. Activation of sympathetic nerves to the kidney ✓**
- D. Increased extracellular fluid volume

Q137. Which hormones is primarily responsible for increasing potassium excretion in urine?

- A. Antidiuretic hormone (ADH)
- B. Parathyroid hormone (PTH)
- C. Aldosterone ✓**
- D. Atrial natriuretic peptide (ANP)

Q138. A patient's creatinine clearance is measured to be 70 mL/min, while the normal average is 125 mL/min. What does this most likely indicate?

- A. Increased kidney function.
- B. Normal kidney function, as creatinine clearance varies widely.
- C. Decreased glomerular filtration rate (GFR). ✓**

D. Excessive hydration.

Q139. Which best defines renal clearance?

- A. The rate at which a substance is reabsorbed by the renal tubules.
- B. The total amount of a substance filtered by the glomeruli per day.
- C. The volume of plasma completely cleared of a substance by the kidneys per unit time.** ✓
- D. The concentration of a substance in the urine relative to its plasma concentration.

Q140. Which is true about acid-base regulation in the kidney?

- A. The kidneys excrete fixed acids but cannot excrete volatile acids.
- B. The primary mechanism is to directly excrete bicarbonate ions to lower pH.
- C. They primarily regulate pH by reabsorbing bicarbonate and secreting hydrogen ions.** ✓
- D. Renal compensation for acidosis is immediate and short-lived.

Q141. A patient's urine sample shows proteinuria and red blood cells under microscopic examination. Which part of the nephron is most likely damaged? 241

- A. Glomerulus** ✓
- B. Proximal convoluted tubule
- C. Loop of Henle
- D. Collecting duct

Q142. In a healthy individual, Which findings is normal in urine?

- A. Glucose
- B. Protein
- C. Red blood cells
- D. Creatinine** ✓

Q143. Which changes would occur in hyperaldosteronism (Conn's syndrome)?

- A. Hyperkalemia and metabolic acidosis
- B. Hypokalemia and metabolic alkalosis** ✓
- C. Hyponatremia and hypotension
- D. Hyperglycemia and hypercalcemia

Q144. If both afferent and efferent arterioles constrict, but the efferent arteriole constricts more strongly, what happens to GFR and renal plasma flow (RPF)?

- A. GFR decreases, RPF increases
- B. GFR increases, RPF increases
- C. GFR decreases, RPF decreases

D. GFR increases, RPF decreases ✓

Q145. What does the clearance of a substance that is freely filtered at the glomerulus, completely secreted by the tubules, and not reabsorbed best represent?

- A. Glomerular Filtration Rate (GFR)
- B. Tubular secretion rate
- C. Renal Plasma Flow (RPF) ✓**
- D. Filtration Fraction

Q146. Which correctly describes the mechanism of renin release from the juxtaglomerular (JG) cells?

- A. Increased blood pressure in the afferent arteriole
- B. Increased sodium delivery to the macula densa
- C. Stimulation of beta-1 adrenergic receptors ✓**
- D. Increased sympathetic tone to the renal medulla

Q147. Which best describes innate immunity?

- A. Highly specific recognition of pathogens
- B. Development of immunological memory after exposure
- C. Immediate, non-specific response to pathogens ✓**
- D. Primarily mediated by T-lymphocytes

Q148. What is the primary feature that distinguishes adaptive immunity from innate immunity?

- A. Rapid onset of response
- B. Physical barrier formation
- C. Immunological memory ✓**
- D. Phagocytic activity

Q149. Which cells are primarily responsible for cell-mediated immunity?

- A. B lymphocytes
- B. Plasma cells
- C. Neutrophils
- D. T lymphocytes ✓**

Q150. Vigorous physical activity (such as exercise or hard labor) increases the body's demand for oxygen. The first-line physiologic response to this demand is an increase in heart rate, breathing rate, and depth of breathing. During heavy exercise, dyspnea will appear when the dyspneic index will fall just below?

- A. 0.7
- B. 0.5
- C. 0.3 ✓**
- D. 0.9

Q151. Which of following energy systems uses almost entirely in muscle metabolism during 100-meter dash?

- A. Oxidative phosphorylation
- B. Glycolysis
- C. Adenosine triphosphate-phosphocreatine (ATP-PC) system ✓**
- D. Beta-oxidation

Q152. Which is not a component of the innate immune system?

- A. Macrophages
- B. Natural killer (NK) cells
- C. B lymphocytes ✓**
- D. Neutrophils

Q153. Which pairs is correctly matched regarding immunity type?

- A. Natural active immunity: Vaccination
- B. Artificial passive immunity: Breastfeeding
- C. Natural passive immunity: Transplacental transfer of antibodies ✓**
- D. Artificial active immunity: Exposure to a pathogen

Q154. Which is the first antibody produced in the primary immune response?

- A. IgA
- B. IgG
- C. IgE
- D. IgM ✓**

Q155. Which is NOT a function of the complement system in enhancing immune defense?

- A. Opsonization
- B. Direct lysis of target cells
- C. Antibody production ✓**
- D. Chemotaxis

Q156. What is the primary function of the immune system?

- A. To regulate blood glucose levels

- B. To produce hormones
- C. To protect the body from pathogens and foreign substances ✓**
- D. To facilitate gas exchange

Q157. What is the term for the ability of the immune system to remember and respond more effectively to a previously encountered pathogen?

- A. Autoimmunity
- B. Immunological tolerance
- C. Immunological memory ✓**
- D. Hypersensitivity

Q158. What type of immune mechanism primarily mediates serum sickness following administration of antiserum?

- A. Type I hypersensitivity
- B. Type II hypersensitivity
- C. Type III hypersensitivity ✓**
- D. Type IV hypersensitivity

Q159. Which part of the brain acts as the principal center for thermoregulation in humans?

- A. Cerebellum
- B. Medulla oblongata
- C. Hypothalamus ✓**
- D. Thalamus

Q160. Which responses is most likely to occur when core body temperature decreases?

- A. Vasodilation of cutaneous blood vessels
- B. Increased sweating
- C. Shivering ✓**
- D. Decreased metabolic rate

Q161. What is the primary mechanism of heat loss at rest under normal environmental conditions?

- A. Radiation ✓**
- B. Evaporation
- C. Conduction
- D. Convection

Q162. Stimulation of the anterior hypothalamus leads to Which physiological effects?

- A. Hypothermia and parasympathetic responses ✓**
- B. Hyperthermia and sympathetic responses
- C. Increased hunger and rage
- D. Decreased heart rate and inhibition of thirst

Q163. During exposure to high environmental temperatures, Which mechanisms helps prevent excessive rise in body temperature?

- A. Cutaneous vasodilation and increased sweat production ✓**
- B. Cutaneous vasoconstriction and decreased metabolic rate
- C. Skeletal muscle shivering and piloerection
- D. Decreased respiratory rate and internal heat storage

Q164. Which describes a behavioral mechanism of thermoregulation?

- A. Shivering to generate internal heat
- B. Sweating to promote evaporative heat loss
- C. Vasoconstriction of cutaneous blood vessels
- D. Putting on a heavy sweater when feeling cold ✓**

Q165. In cold exposure, which hormonal response enhances long-term heat production?

- A. Increased secretion of thyroid hormones (T3 and T4) ✓**
- B. Decreased secretion of insulin
- C. Increased secretion of atrial natriuretic peptide
- D. Decreased secretion of antidiuretic hormone

Q166. Which is a local cutaneous thermal reflex?

- A. Axon reflex vasodilation ✓**
- B. Shivering thermogenesis
- C. Sweating via sympathetic cholinergic activation
- D. Central hypothalamic thermoregulation

Q167. A runner experiences profuse sweating but minimal heat loss due to high humidity. What mechanism explains this?

- A. Reduction of the vapor pressure gradient between the skin and the environment ✓**
- B. Inhibition of sympathetic cholinergic nerve fibers innervating sweat glands
- C. Direct vasoconstriction of cutaneous blood vessels
- D. Decreased production of eccrine sweat gland secretion

Q168. Which correctly describes heat stroke?

- A. A core body temperature above 40°C accompanied by central nervous system dysfunction ✓**
- B. A mild form of heat illness characterized by profuse sweating and normal mental status
- C. A condition primarily caused by salt depletion without significant water loss
- D. A drop in core body temperature below 35°C due to prolonged exposure to cold

Q169. Which mechanisms is most important in preventing a dangerous fall in core temperature during prolonged cold exposure?

- A. Shivering ✓**
- B. Cutaneous vasodilation
- C. Decreased thyroid hormone secretion
- D. Sweating

Q170. During fever, Which events occurs first following pyrogen exposure?

- A. Release of endogenous pyrogens (cytokines) into the circulation
- B. Increased heat production via shivering and increased metabolism
- C. Release of prostaglandins (specifically PGE2) in the preoptic area of the hypothalamus ✓**
- D. Cutaneous vasoconstriction to decrease heat loss

Q171. Which part of the brain is primarily responsible for maintenance of posture and equilibrium?

- A. Cerebrum
- B. Cerebellum ✓**
- C. Hypothalamus
- D. Medulla oblongata

Q172. How do the basal ganglia contribute to motor control?

- A. Planning and modulating movement patterns by inhibiting unwanted motor activity and facilitating desired movements ✓**
- B. Directly initiating voluntary motor commands from the primary motor cortex to spinal motor neurons
- C. Detecting movement errors and coordinating rapid, online adjustments during ongoing muscle contraction
- D. Relaying sensory information from peripheral mechanoreceptors to the somatosensory cortex

Q173. What is the primary function of the vestibulocerebellum (flocculonodular lobe)?

- A. Regulation of balance and eye movements ✓**

- B. Planning and initiation of voluntary limb movements
- C. Execution of rapid sequential multi-joint movements
- D. Control of proximal muscles for posture and locomotion

Q174. Which structures is most important for maintaining muscle tone?

- A. Muscle spindles ✓**
- B. Golgi tendon organs
- C. Pacinian corpuscles
- D. Motor end plates

Q175. Damage to the cerebellum causes Which characteristic features?

- A. Ataxia, intention tremor, and dysmetria ✓**
- B. Spastic paralysis, hyperreflexia, and Babinski sign
- C. Resting tremor, rigidity, and bradykinesia
- D. Flaccid paralysis, muscle atrophy, and fasciculations

Q176. Which pathways is most important in the maintenance of balance by adjusting head and body position relative to gravity?

- A. Vestibulospinal tract ✓**
- B. Corticospinal tract
- C. Rubrospinal tract
- D. Tectospinal tract

Q177. Which concept best describes the maintenance of a stable internal environment in the human body?

- A. Homeostasis ✓**
- B. Hemopoiesis
- C. Hyperemia
- D. Hemostasis

Q178. Which is an example of a negative feedback mechanism?

- A. Oxytocin release enhancing uterine contractions during childbirth
- B. Platelet plug formation during blood clotting
- C. Blood pressure regulation via the baroreceptor reflex ✓**
- D. The surge of luteinizing hormone (LH) triggered by estrogen before ovulation

Q179. Which system acts as the rapid control system for maintaining internal environment stability?

- A. Endocrine system
- B. Nervous system ✓**

- C. Lymphatic system
- D. Integumentary system

Q180. Which statements regarding somatic sensations is true?

- A. Pain and temperature information is primarily carried by the dorsal column-medial lemniscal pathway.
- B. Rapidly adapting receptors, such as Pacinian corpuscles, are well-suited for detecting sustained pressure and steady touch.
- C. The primary somatosensory cortex (S1) is located in the postcentral gyrus of the parietal lobe. ✓**
- D. Fine touch, vibration, and conscious proprioception cross the midline at the level of the spinal cord.

Q181. Which sensory receptors is responsible for detecting fine touch and texture?

- A. Pacinian corpuscles
- B. Meissner's corpuscles ✓**
- C. Ruffini endings
- D. Free nerve endings

Q182. The pain pathway primarily involves Which tracts?

- A. Spinothalamic tract ✓**
- B. Dorsal column-medial lemniscal pathway
- C. Corticospinal tract
- D. Tectospinal tract

Q183. Which neurotransmitter is most associated with the transmission of pain in the spinal cord?

- A. Acetylcholine
- B. Substance P ✓**
- C. GABA
- D. Dopamine

Q184. Pain due to tissue injury and inflammation is mainly mediated by which substances?

- A. Prostaglandins, bradykinin, and substance P ✓**
- B. Acetylcholine, GABA, and dopamine
- C. Insulin, glucagon, and somatostatin
- D. Thyroxine, calcitonin, and parathyroid hormone

Q185. Which is true about headache caused by tension or muscle contraction?

A. It is typically bilateral, dull, aching, and non-pulsatile. ✓

- B. It is strictly unilateral and throbbing in character.
- C. It is associated with visual aura and nausea.
- D. It is caused by intracranial vasodilation.

Q186. What is the main stimulus responsible for increased ventilation at high altitude?

A. Hypoxemia acting on peripheral chemoreceptors ✓

- B. Hypercapnia acting on central chemoreceptors
- C. Increased blood pH acting on the carotid bodies
- D. Increased metabolic rate acting on temperature receptors

Q187. Which acute physiological changes occurs during ascent to high altitude?

A. Increased hyperventilation leading to respiratory alkalosis ✓

- B. Increased synthesis of 2,3-DPG in red blood cells
- C. Decreased resting heart rate to conserve oxygen
- D. Metabolic alkalosis due to renal bicarbonate retention

Q188. Chronic exposure to high altitude leads to Which adaptations?

A. Increased erythropoiesis mediated by elevated erythropoietin ✓

- B. Decreased 2,3-BPG concentration in red blood cells
- C. Metabolic alkalosis due to sustained hyperventilation
- D. Decreased pulmonary capillary density

Q189. Which best explains why females generally have a lower basal metabolic rate than males?

A. Females have a lower percentage of lean muscle mass relative to total body fat compared to males ✓

- B. Females have lower circulating levels of thyroid hormones at all times
- C. Females have a lower core body temperature that decreases metabolic enzyme kinetics
- D. Females have a higher surface-area-to-mass ratio, leading to greater heat loss

Q190. During moderate exercise, Which changes occurs?

A. Increased skeletal muscle blood flow ✓

- B. Decreased cardiac output
- C. Constriction of skeletal muscle arterioles
- D. Decreased respiratory rate

Q191. What is the primary action of leptin, a hormone secreted by adipocytes?

- A. Stimulates appetite and decreases energy expenditure
- B. Suppresses appetite and increases energy expenditure ✓**
- C. Increases glucose uptake in skeletal muscle
- D. Promotes fat storage in white adipose tissue

Q192. At high altitude, after several days of acclimatization, arterial oxygen content remains subnormal despite increased ventilation. Which best explains this finding?

- A. Hyperventilation causes an increased loss of CO₂, leading to respiratory alkalosis which shifts the oxygen-hemoglobin dissociation curve to the left, and full normalization of arterial PO₂ is limited by the ongoing low barometric pressure. ✓**
- B. Renal compensation for respiratory alkalosis leads to an increased production of 2,3-BPG, which decreases hemoglobin's affinity for oxygen and prevents full saturation.
- C. Pulmonary vasoconstriction induced by alveolar hypoxia leads to severe ventilation-perfusion mismatch that completely prevents oxygen uptake in the pulmonary capillaries.
- D. Downregulation of erythropoietin receptors on bone marrow progenitor cells limits the increase in total hemoglobin mass.

Q193. During chronic exposure to altitude, Which contributes most to the increase in oxygen delivery to tissues?

- A. Increased erythropoiesis resulting in elevated hemoglobin concentration ✓**
- B. Permanent leftward shift of the oxygen-hemoglobin dissociation curve
- C. Sustained maximal hyperventilation leading to respiratory alkalosis
- D. Decreased tissue capillary density

Q194. With aging, Which parameters increases rather than decreases?

- A. Pulse pressure ✓**
- B. Cardiac output at rest
- C. Vital capacity
- D. Glomerular filtration rate

Q195. Which equations correctly represents the relationship between blood flow, pressure, and resistance?

- A. $\text{Flow} = (\text{Pressure Gradient}) / \text{Resistance}$ ✓**
- B. $\text{Flow} = \text{Resistance} / (\text{Pressure Gradient})$
- C. $\text{Flow} = \text{Pressure Gradient} \times \text{Resistance}$
- D. $\text{Flow} = \text{Pressure Gradient} + \text{Resistance}$

Q196. According to Poiseuille's law, the resistance to laminar blood flow is inversely proportional to Which?

- A. Viscosity of the blood

- B. Length of the blood vessel
- C. Radius of the blood vessel to the fourth power ✓**
- D. Pressure gradient across the vessel

Q197. By how much does the resistance to blood flow increase if the radius of a blood vessel is reduced to half of its original value?

- A. 2-fold
- B. 4-fold
- C. 8-fold
- D. 16-fold ✓**

Q198. Which is the most important short-term mechanism regulating arterial blood pressure?

- A. Renin-angiotensin-aldosterone system
- B. Baroreceptor reflex ✓**
- C. Fluid intake regulation
- D. Capillary fluid shift

Q199. Which will occur immediately if arterial blood pressure suddenly decreases?

- A. Increased sympathetic nerve activity ✓**
- B. Decreased heart rate
- C. Vasodilation of systemic arterioles
- D. Decreased renin release

Q200. Which factors will decrease blood viscosity and therefore increase blood flow?

- A. Anemia and decreased hematocrit ✓**
- B. Polycythemia and dehydration
- C. Vasoconstriction and increased hematocrit
- D. Increased plasma protein concentration

Q201. Which events primarily terminates the plateau phase of the ventricular action potential?

- A. Inactivation of L-type calcium channels and activation of delayed rectifier potassium channels ✓**
- B. Activation of fast sodium channels and inhibition of the sodium-calcium exchanger
- C. Closure of inward rectifier potassium channels and opening of funny sodium channels
- D. Phosphorylation of phospholamban and rapid uptake of calcium by the sarcoplasmic reticulum

Q202. Which is the primary source of calcium that triggers contraction in cardiac muscle?

- A. Extracellular fluid via L-type calcium channels ✓**
- B. Sarcoplasmic reticulum via IP3 receptors
- C. Mitochondria via sodium-calcium exchangers
- D. Caveolae via voltage-gated T-type channels

Q203. Which physiological relationships is explained by the Frank–Starling mechanism of the heart?

- A. Stroke volume and end-diastolic volume ✓**
- B. Heart rate and cardiac output
- C. Mean arterial pressure and total peripheral resistance
- D. Coronary blood flow and myocardial oxygen consumption

Q204. Which ion channel is primarily responsible for the Phase 4 spontaneous depolarization in pacemaker cells of the sinoatrial (SA) node?

- A. L-type calcium channel
- B. T-type calcium channel
- C. Funny current (If) channel ✓**
- D. Inward rectifier potassium channel

Q205. Which best describes the sequence of mechanical events during isovolumetric relaxation?

- A. All heart valves are closed; ventricular pressure decreases rapidly while ventricular volume remains constant. ✓**
- B. The aortic and pulmonary valves are open while the AV valves are closed, and ventricular volume decreases.
- C. The AV valves are open while the semilunar valves are closed, and ventricular volume increases.
- D. All heart valves are open, and blood flows passively through the ventricles.

Q206. Which features distinguishes excitation–contraction coupling in cardiac muscle from that in skeletal muscle?

- A. Calcium-induced calcium release via ryanodine receptors ✓**
- B. Direct mechanical coupling between L-type calcium channels and ryanodine receptors
- C. Presence of troponin C as the calcium receptor
- D. Depolarization of the transverse tubules

Q207. Which statements best explains why cardiac muscle cannot undergo tetanic contraction?

A. The absolute refractory period of cardiac muscle is nearly as long as the entire duration of mechanical contraction, preventing a new action potential from occurring during the contraction phase. ✓

B. The sarcoplasmic reticulum in cardiac muscle lacks the capacity to reaccumulate calcium ions rapidly enough between successive stimuli.

C. Gap junctions between cardiac muscle cells allow continuous electrical activity that inherently prevents relaxation.

D. Vagus nerve stimulation continuously hyperpolarizes the pacemaker and working myocardial cells, preventing summation of twitches.

Q208. Which autonomic effects will cause an increase in heart rate?

A. Sympathetic stimulation and parasympathetic inhibition ✓

B. Parasympathetic stimulation and sympathetic inhibition

C. Parasympathetic stimulation alone

D. Decreased circulating catecholamines and increased acetylcholine

Q209. Which is the most important immediate determinant of cardiac output in a healthy individual?

A. Venous return (preload) ✓

B. Total peripheral resistance

C. Heart rate

D. Myocardial contractility

Q210. Which increases would lead to a decrease in cardiac output?

A. End-systolic volume ✓

B. Heart rate

C. Myocardial contractility

D. End-diastolic volume

Q211. Which mechanisms best explains the Frank–Starling law of the heart?

A. Increased venous return stretches the ventricular myocardium, optimizing the overlap of actin and myosin filaments and increasing calcium sensitivity. ✓

B. Increased sympathetic stimulation enhances the rate of calcium uptake into the sarcoplasmic reticulum during diastole.

C. Decreased afterload allows for a more complete emptying of the ventricle during systole, independent of fiber length.

D. Hyperpolarization of the sinoatrial node increases the rate of phase 4 depolarization, leading to a higher stroke volume.

Q212. Which statements regarding baroreceptor reflex is true?

- A. Increased arterial pressure stretches the carotid sinus, leading to increased parasympathetic outflow to the heart and decreased sympathetic tone to blood vessels.
- B. Activation of the baroreceptor reflex during hemorrhage causes reflex tachycardia and peripheral vasodilation.

C. The carotid sinus baroreceptors send afferent signals to the brainstem primarily via the glossopharyngeal nerve (CN IX). ✓

- D. Baroreceptors are most sensitive to absolute changes in mean arterial pressure rather than the rate of pressure change.

Q213. Which statements about cardiac output regulation is true?

A. Stroke volume is primarily determined by preload, afterload, and myocardial contractility. ✓

- B. Parasympathetic stimulation significantly increases the heart rate and force of contraction.
- C. Cardiac output decreases when venous return increases due to the Frank-Starling mechanism.
- D. Afterload is defined as the end-diastolic volume when the ventricle is fully filled.

Q214. Which immediate cardiovascular responses occurs when a person suddenly stands up from a supine position due to activation of the baroreceptor reflex?

A. Increased heart rate and vasoconstriction ✓

- B. Decreased heart rate and vasodilation
- C. Increased heart rate and vasodilation
- D. Decreased heart rate and vasoconstriction

Q215. Which changes most effectively raises mean arterial pressure in the short term?

A. Increased sympathetic nerve activity, increased heart rate, and vasoconstriction ✓

- B. Decreased sympathetic nerve activity, decreased heart rate, and vasodilation
- C. Increased parasympathetic nerve activity, decreased heart rate, and vasoconstriction
- D. Decreased parasympathetic nerve activity, increased heart rate, and vasodilation

Q216. Which factors can still increase cardiac output during exercise in a patient with a completely denervated heart after heart transplantation?

A. Increased venous return (preload) and circulating catecholamines ✓

- B. Parasympathetic nervous system stimulation and acetylcholine release
- C. Direct sympathetic nerve stimulation from the cardiac plexus
- D. Baroreceptor reflex-mediated changes in intrinsic heart rate

Q217. Which vessels are known as the resistance vessels of the circulatory system?

- A. Large arteries

B. Arterioles ✓

C. Capillaries

D. Venuels

Q218. Which vessels are considered the capacitance vessels of the circulatory system?

A. Arterioles

B. Capillaries

C. Veins ✓

D. Aorta

Q219. Which factor has the greatest influence on blood flow through a vessel according to Poiseuille's law?

A. Vessel radius ✓

B. Blood viscosity

C. Vessel length

D. Blood pressure gradient

Q220. Which type of vessel shows the highest total cross-sectional area and the lowest mean velocity of blood flow?

A. Aorta

B. Large arteries

C. Arterioles

D. Capillaries ✓

Q221. According to Laplace's law, which type of blood vessel is least likely to rupture despite having a very thin wall?

A. Capillaries ✓

B. Aorta

C. Vena cava

D. Muscular arteries

Q222. Which segment of the systemic circulation contributes the most to total peripheral resistance (TPR)?

A. Aorta

B. Large arteries

C. Arterioles ✓

D. Capillaries

Q223. Which muscles is primarily responsible for quiet inspiration at rest?

A. Diaphragm ✓

- B. Internal intercostal muscles
- C. Abdominal muscles
- D. Sternocleidomastoid

Q224. Which statements best describes normal expiration at rest?

- A. It is an active process driven by contraction of the internal intercostal muscles.
- B. It is a passive process driven by the elastic recoil of the lungs and chest wall. ✓**
- C. It requires active contraction of the diaphragm and abdominal muscles.
- D. It involves a decrease in intra-alveolar pressure below atmospheric pressure.

Q225. Which pressures is always negative during normal quiet breathing?

- A. Intrapleural pressure ✓**
- B. Alveolar pressure
- C. Intra-alveolar pressure
- D. Atmospheric pressure

Q226. Which factor serves as the primary physiological stimulus for respiration under normal conditions?

- A. Increased arterial partial pressure of carbon dioxide (PaCO₂) ✓**
- B. Decreased arterial partial pressure of oxygen (PaO₂)
- C. Increased arterial blood pH
- D. Decreased arterial bicarbonate concentration

Q227. Which statements about surfactant is correct?

- A. It is secreted by type II alveolar cells and decreases surface tension. ✓**
- B. It is composed primarily of carbohydrates and increases lung compliance.
- C. It is secreted by type I alveolar cells and increases surface tension.
- D. It is produced by club cells and stabilizes small alveoli by increasing their collapsing pressure.

Q228. Which best describes the relationship between alveolar ventilation and arterial PCO₂?

- A. Inverse proportional relationship (hyperventilation decreases arterial PCO₂) ✓**
- B. Direct proportional relationship (hyperventilation increases arterial PCO₂)
- C. Exponential relationship (ventilation equals the square of arterial PCO₂)
- D. No relationship, as arterial PCO₂ is strictly determined by cardiac output

Q229. Which events contributes most to the decrease in intrapleural pressure during quiet inspiration?

A. Contraction of the diaphragm leading to an increase in vertical dimension of the thoracic cavity ✓

- B. Relaxation of the internal intercostal muscles
- C. Active pumping of gas out of the intrapleural space
- D. Increased surface tension of the alveolar fluid

Q230. Which statements correctly explains why the apex of the lung has a higher ventilation-perfusion (V/Q) ratio than the base?

A. Gravity causes greater relative increases in both ventilation and perfusion at the base, but perfusion increases to a much greater extent. ✓

- B. Gravity causes greater relative increases in both ventilation and perfusion at the apex, but ventilation increases to a much greater extent.
- C. The apex has lower compliance, preventing adequate ventilation despite high perfusion.
- D. The base is compressed by abdominal contents, reducing its ventilation to near zero.

Q231. Which statement best explains the Bohr effect in systemic capillaries?

A. An increase in carbon dioxide and hydrogen ion concentration shifts the oxygen-hemoglobin dissociation curve to the right, facilitating oxygen unloading in tissues. ✓

- B. A decrease in temperature shifts the oxygen-hemoglobin dissociation curve to the right, enhancing oxygen release to metabolically active cells.
- C. An increase in pH due to lactic acid accumulation enhances the affinity of hemoglobin for oxygen in systemic capillaries.
- D. A decrease in 2,3-BPG concentration in systemic capillaries causes hemoglobin to release more oxygen to the surrounding tissues.

Q232. Which lung volumes cannot be measured by a simple spirometer?

A. Residual volume, functional residual capacity, and total lung capacity ✓

- B. Tidal volume, inspiratory reserve volume, and expiratory reserve volume
- C. Vital capacity, inspiratory capacity, and tidal volume
- D. Expiratory reserve volume, residual volume, and vital capacity

Q233. Which conditions does not cause a rightward shift of the oxygen-hemoglobin dissociation curve?

- A. Increased body temperature
- B. Decreased blood pH
- C. Increased 2,3-BPG
- D. Decreased arterial partial pressure of carbon dioxide ✓**

Q234. What is the approximate percentage saturation of hemoglobin with oxygen at a PO_2 of about 40 mm Hg?

- A. 25%
- B. 50%
- C. 75% ✓**
- D. 100%

Q235. Which statements about oxygen transport in blood is correct?

- A. Most of the oxygen in the blood is transported bound to hemoglobin ✓**
- B. Carbon monoxide decreases the affinity of hemoglobin for oxygen
- C. Oxygen transport is independent of blood pH
- D. A decrease in temperature decreases the affinity of hemoglobin for oxygen

Q236. Which factors mainly explains why alveolar PO_2 (PAO_2) is lower than the inspired PO_2 ?

- A. Continuous uptake of O_2 by pulmonary capillaries and addition of water vapor and CO_2 in the alveoli ✓**
- B. High residual volume in the anatomical dead space
- C. Active transport of oxygen out of the alveoli by alveolar macrophages
- D. Diffusion limitation of oxygen across the respiratory membrane

Q237. Which conditions leads to an increase in the alveolar–arterial oxygen gradient ($A-a$ gradient)?

- A. Right-to-left shunt ✓**
- B. Hypoventilation
- C. High altitude
- D. Normal aging

Q238. What is the approximate oxygen consumption ($\dot{V}O_2$) per minute in a healthy adult at rest?

- A. 50 mL/min
- B. 125 mL/min
- C. 250 mL/min ✓**
- D. 500 mL/min

Q239. Which form of CO_2 transport accounts for the largest percentage of total CO_2 carried in blood?

- A. Dissolved CO_2 gas
- B. Carbaminohemoglobin
- C. Bicarbonate ions ✓**
- D. Carboxyhemoglobin

Q240. Which factors determines oxygen utilization coefficient (O₂ extraction ratio)?

A. Ratio of oxygen consumption to oxygen delivery ✓

- B. Arterial oxygen partial pressure alone
- C. Total hemoglobin concentration alone
- D. Cardiac output multiplied by heart rate

Q241. Which region of the brainstem contains the primary respiratory rhythm generator?

A. Pons

B. Medulla oblongata ✓

- C. Midbrain
- D. Cerebellum

Q242. Which primarily stimulates the central chemoreceptors that control respiration?

A. Decreased pH of the cerebrospinal fluid ✓

- B. Decreased arterial partial pressure of oxygen
- C. Increased arterial partial pressure of carbon dioxide
- D. Increased blood pH

Q243. Which depresses respiratory centers and may cause respiratory arrest in overdose?

A. Opioids ✓

- B. Amphetamines
- C. Cocaine
- D. Nicotine

Q244. Which statements about Hering–Breuer reflex is correct?

A. It is mediated by pulmonary stretch receptors that limit the depth of inspiration. ✓

- B. It is initiated by peripheral chemoreceptors responding to hypoxia.
- C. It stimulates the apneustic center to prolong inspiration.
- D. It causes bronchoconstriction via activation of parasympathetic C-fibers.

Q245. Which mechanisms primarily causes the initial increase in ventilation during hypoxia at high altitude?

A. Stimulation of peripheral chemoreceptors by decreased arterial PO₂ ✓

- B. Stimulation of central chemoreceptors by decreased arterial PCO₂
- C. Direct depression of the respiratory center by hypoxia
- D. Activation of pulmonary stretch receptors via the Hering-Breuer reflex

Q246. During breath-holding, Which changes is the primary driver that eventually forces a person to resume breathing?

- A. Decrease in arterial blood pH
- B. Decrease in arterial blood oxygen tension
- C. Increase in arterial blood carbon dioxide tension ✓**
- D. Increase in arterial blood bicarbonate concentration

Q247. A climber at high altitude hyperventilates. After a few days, ventilation remains high despite low arterial PCO₂. Which mechanism best explains this adaptation?

- A. Renal excretion of bicarbonate decreases plasma pH around the brain, stimulating central chemoreceptors ✓**
- B. Direct stimulation of carotid bodies by severe hypoxia overrides the inhibitory effect of hypocapnia
- C. Increased sensitivity of peripheral chemoreceptors to arterial PO₂ overrides central inhibition
- D. Upregulation of carbonic anhydrase in the medulla increases local PCO₂

Q248. Which best describes the role of the apneustic center in the pons?

- A. Promotes inspiration by stimulating the medullary inspiratory neurons and prolonging inhalation ✓**
- B. Inhibits the medullary inspiratory neurons to terminate inspiration and control respiratory rate
- C. Acts as the primary central chemoreceptor responding to changes in arterial blood pH
- D. Detects arterial blood oxygen levels and drives the hypoxic ventilatory response

Q249. Which describes the Hering–Breuer inflation reflex?

- A. Pulmonary stretch receptors terminate inspiration and prevent overinflation of the lungs ✓**
- B. Chemoreceptors in the carotid body stimulate ventilation in response to hypoxemia
- C. Irritant receptors in the airways cause bronchoconstriction and coughing
- D. Central chemoreceptors respond to decreases in cerebrospinal fluid pH

Q250. Which hormones acts through intracellular receptors that directly regulate gene transcription?

- A. Cortisol ✓**
- B. Insulin
- C. Epinephrine
- D. Glucagon

Q251. Which is a second messenger used by many peptide hormones such as vasopressin and epinephrine (acting on α receptors)?

- A. Cyclic AMP
- B. Inositol 1,4,5-trisphosphate (IP3) and Diacylglycerol (DAG) ✓**
- C. cGMP
- D. Receptor tyrosine kinase auto-phosphorylation

Q252. Which anterior pituitary hormones acts via JAK–STAT (tyrosine kinase–associated) signaling pathway?

- A. Growth hormone ✓**
- B. Thyroid-stimulating hormone
- C. Adrenocorticotrophic hormone
- D. Luteinizing hormone

Q253. Which pituitary hormones is secreted by the posterior pituitary and not synthesized there?

- A. Oxytocin ✓**
- B. Growth hormone
- C. Prolactin
- D. Thyroid-stimulating hormone

Q254. Which correctly describes the feedback relationship between the hypothalamus, pituitary, and thyroid glands?

- A. Thyrotropin-releasing hormone (TRH) from the hypothalamus stimulates thyroid-stimulating hormone (TSH) release from the anterior pituitary, which then stimulates thyroid hormone production; thyroid hormones subsequently exert negative feedback on both the hypothalamus and anterior pituitary. ✓**
- B. Thyroid hormones exert positive feedback on the hypothalamus to increase TRH secretion when metabolic rate is high.
- C. Thyroid-stimulating hormone (TSH) from the posterior pituitary directly regulates the secretion of TRH from the hypothalamus via a short-loop positive feedback mechanism.
- D. Thyroid hormones inhibit the release of TSH from the anterior pituitary but stimulate the hypothalamus to release more TRH in a balanced feed-forward loop.

Q255. Which accurately describes calcitonin's physiological role?

- A. It lowers plasma calcium levels by inhibiting osteoclast activity and decreasing renal calcium reabsorption. ✓**
- B. It raises plasma calcium levels by stimulating osteoblast activity and increasing intestinal calcium absorption.
- C. It stimulates bone resorption to increase serum calcium levels during periods of hypocalcemia.
- D. It promotes the hydroxylation of 25-hydroxyvitamin D to its active form in the kidneys.

Q256. Which statements best explains the mechanism of steroid hormone action?

- A. Binding to intracellular receptors and altering gene transcription ✓**
- B. Binding to G-protein coupled cell surface receptors to activate adenylate cyclase
- C. Activating receptor tyrosine kinases through dimerization and autophosphorylation
- D. Opening ligand-gated ion channels to directly alter membrane potential

Q257. Which pituitary cell types produces a hormone whose secretion is primarily under inhibitory hypothalamic control?

- A. Somatotrophs
- B. Thyrotrophs
- C. Lactotrophs ✓**
- D. Corticotrophs

Q258. Which hormones is glycoprotein in structure and shares an identical α -subunit with others?

- A. Insulin, Growth Hormone, and Prolactin
- B. Thyroid-Stimulating Hormone (TSH), Luteinizing Hormone (LH), Follicle-Stimulating Hormone (FSH), and human Chorionic Gonadotropin (hCG) ✓**
- C. Adrenocorticotrophic Hormone (ACTH), Melanocyte-Stimulating Hormone (MSH), and Oxytocin
- D. Cortisol, Aldosterone, and DHEA

Q259. A patient presents with polyuria, polydipsia, and hypernatremia. Urine osmolality fails to increase after water deprivation but rises significantly after vasopressin injection. What is the most likely diagnosis?

- A. Central diabetes insipidus ✓**
- B. Nephrogenic diabetes insipidus
- C. Primary polydipsia
- D. Syndrome of inappropriate antidiuretic hormone secretion

Q260. Which describes the principal action of parathyroid hormone (PTH) on the kidney?

- A. Increases calcium reabsorption and decreases phosphate reabsorption ✓**
- B. Increases both calcium and phosphate reabsorption
- C. Decreases calcium reabsorption and increases phosphate reabsorption
- D. Decreases both calcium and phosphate reabsorption

Q261. A patient has muscle weakness, bone pain, and elevated plasma calcium with low phosphate. Which condition is most consistent with these findings?

A. Primary hyperparathyroidism ✓

- B. Vitamin D deficiency
- C. Chronic kidney disease
- D. Hypoparathyroidism

Q262. Which best stimulates glucocorticoid secretion from the adrenal cortex?

A. Adrenocorticotrophic hormone (ACTH) ✓

- B. Growth hormone (GH)
- C. Antidiuretic hormone (ADH)
- D. Angiotensin II

Q263. Which is a physiological effect of glucocorticoids (cortisol)?

A. Stimulation of hepatic gluconeogenesis ✓

- B. Inhibition of lipolysis in adipose tissue
- C. Stimulation of glucose uptake in skeletal muscle
- D. Inhibition of protein degradation in muscle

Q264. Which factors is the major regulator of aldosterone secretion under normal physiological conditions?

A. Angiotensin II and extracellular potassium ion concentration ✓

- B. Adrenocorticotrophic hormone (ACTH) and plasma sodium levels
- C. Atrial natriuretic peptide (ANP) and blood pressure
- D. Sympathetic nervous system activity and plasma calcium levels

Q265. Which correctly describes the zones of the adrenal cortex and their primary hormone products?

A. Zona glomerulosa produces mineralocorticoids, zona fasciculata produces glucocorticoids, and zona reticularis produces androgens. ✓

- B. Zona glomerulosa produces glucocorticoids, zona fasciculata produces mineralocorticoids, and zona reticularis produces catecholamines.
- C. Zona glomerulosa produces androgens, zona fasciculata produces catecholamines, and zona reticularis produces glucocorticoids.
- D. Zona glomerulosa produces mineralocorticoids, zona fasciculata produces androgens, and zona reticularis produces glucocorticoids.

Q266. Which hormonal patterns is expected in a patient with Cushing's syndrome caused by a cortisol-secreting adrenal tumor?

A. Increased cortisol, decreased ACTH, decreased CRH ✓

- B. Increased cortisol, increased ACTH, increased CRH
- C. Decreased cortisol, increased ACTH, increased CRH

D. Decreased cortisol, decreased ACTH, increased CRH

Q267. Which enzyme converts norepinephrine → epinephrine in adrenal medulla?

A. Phenylethanolamine N-methyltransferase (PNMT) ✓

B. Tyrosine hydroxylase

C. Dopa decarboxylase

D. Dopamine beta-hydroxylase

Q268. Which receptor mediates most of the metabolic effects of epinephrine such as glycogenolysis and lipolysis?

A. Alpha-1 adrenergic receptor

B. Beta-1 adrenergic receptor

C. Beta-2 adrenergic receptor ✓

D. Alpha-2 adrenergic receptor

Q269. Which mechanisms primarily stimulates catecholamine release from the adrenal medulla during acute stress?

A. Activation of preganglionic sympathetic fibers releasing acetylcholine ✓

B. Direct stimulation by adrenocorticotrophic hormone (ACTH)

C. Decreased blood pressure sensed by renal baroreceptors

D. Parasympathetic nervous system activation

Q270. Which condition results from hypersecretion of aldosterone (primary hyperaldosteronism)?

A. Hyperkalemia

B. Hypokalemia ✓

C. Hyponatremia

D. Hypoglycemia

Q271. Which explains why pheochromocytoma causes paroxysmal hypertension and tachycardia?

A. Episodic oversecretion of catecholamines (epinephrine and norepinephrine) stimulating alpha-1 and beta-1 adrenergic receptors ✓

B. Continuous suppression of the renin-angiotensin-aldosterone system leading to volume-dependent vasoconstriction

C. Autosomal dominant activation of central baroreceptors causing increased sympathetic outflow

D. Massive release of acetylcholine within the adrenal cortex activating nicotinic receptors on vascular smooth muscle

Q272. Which is the rate-limiting step in adrenal steroid hormone synthesis stimulated by ACTH?

- A. Conversion of cholesterol to pregnenolone by cholesterol desmolase (CYP11A1) ✓**
- B. Conversion of progesterone to 11-deoxycorticosterone
- C. Conversion of 11-deoxycortisol to cortisol
- D. Conversion of testosterone to estradiol by aromatase

Q273. Which enzyme deficiency leads to accumulation of 11-deoxycorticosterone (DOC) causing hypertension with low cortisol levels?

- A. 11-beta-hydroxylase deficiency ✓**
- B. 21-hydroxylase deficiency
- C. 17-alpha-hydroxylase deficiency
- D. 3-beta-hydroxysteroid dehydrogenase deficiency

Q274. Which event is primarily triggered by the mid-cycle surge of luteinizing hormone (LH)?

- A. Rupture of the mature ovarian follicle and release of the secondary oocyte ✓**
- B. Initiation of the primary oocyte's first meiotic division
- C. Inhibition of progesterone production by the corpus luteum
- D. Proliferation of the functional layer of the endometrium

Q275. Which hormone predominantly maintains the secretory phase of the endometrium during the menstrual cycle?

- A. Estrogen
- B. Progesterone ✓**
- C. Luteinizing hormone
- D. Follicle-stimulating hormone

Q276. Which day of a normal 28-day menstrual cycle is the luteinizing hormone (LH) surge most likely to occur?

- A. Day 7
- B. Day 14 ✓**
- C. Day 21
- D. Day 28

Q277. Which statement best describes the timing of oogenesis in the human female?

- A. Oogenesis begins during embryonic development, pauses at prophase I until puberty, and completes only upon fertilization. ✓**

- B. Oogenesis begins at puberty, and one oocyte completes meiosis each month until menopause.
- C. Oogenesis begins during embryonic development, pauses at metaphase II, and completes at ovulation.
- D. Oogenesis begins at birth, pauses at prophase I, and completes at each monthly ovulation.

Q278. Which is the major source of progesterone during the first trimester of pregnancy?

- A. Corpus luteum ✓**
- B. Placenta
- C. Anterior pituitary
- D. Adrenal cortex

Q279. Which placental hormones has an action most similar to growth hormone and contributes to maternal insulin resistance during pregnancy?

- A. Human chorionic gonadotropin
- B. Human placental lactogen ✓**
- C. Progesterone
- D. Estriol

Q280. At which stage of meiosis is the secondary oocyte arrested until fertilization occurs?

- A. Prophase I
- B. Metaphase II ✓**
- C. Anaphase I
- D. Telophase II

Q281. During the mid-cycle surge, estradiol exerts positive feedback on the hypothalamic–pituitary axis. Which best explains this physiological event?

- A. Sustained high levels of estradiol switch the responsiveness of the hypothalamic-pituitary axis from negative to positive feedback, dramatically increasing GnRH and LH secretion. ✓**
- B. Estradiol directly suppresses progesterone receptors in the anterior pituitary, leading to an unopposed release of luteinizing hormone.
- C. Low-frequency pulsatile GnRH release induced by rising estradiol inhibits FSH while simultaneously stimulating a massive surge of LH.
- D. Estradiol acts via neural pathways to inhibit the release of endogenous opioids in the hypothalamus, which normally suppress GnRH.

Q282. If the corpus luteum fails to form after ovulation, which hormonal change occurs first?

- A. Decrease in progesterone secretion ✓**

- B. Decrease in follicle-stimulating hormone (FSH)
- C. Increase in luteinizing hormone (LH)
- D. Increase in gonadotropin-releasing hormone (GnRH)

Q283. Which most accurately explains the mechanism of menstruation?

- A. Withdrawal of progesterone and estrogen due to corpus luteum regression leads to endometrial ischemia, prostaglandin release, and sloughing of the functional layer. ✓**
- B. A sudden surge in luteinizing hormone directly induces necrosis of the endometrial functional layer at the end of the luteal phase.
- C. High levels of sustained progesterone stimulate continuous endometrial proliferation until nutrient depletion causes tissue shedding.
- D. Continuous estrogen stimulation without progesterone breakthrough causes endometrial apoptosis and immediate menstrual bleeding.

Q284. Which best describes the feedback pattern during the late follicular phase of the menstrual cycle?

- A. Estrogen exerts negative feedback on FSH and LH secretion
- B. Progesterone exerts positive feedback on the anterior pituitary
- C. High levels of estrogen exert positive feedback on LH and FSH secretion ✓**
- D. Inhibin exerts positive feedback to increase GnRH release

Q285. Which placental hormone is responsible for maintaining maternal insulin resistance and lipolysis, ensuring nutrient availability to the fetus?

- A. Human chorionic gonadotropin (hCG)
- B. Human placental lactogen (hPL) ✓**
- C. Progesterone
- D. Estriol

Q286. Which endocrine change occurs earliest after fertilization?

- A. Rise in human chorionic gonadotropin (hCG)
- B. Surge in luteinizing hormone (LH)
- C. Increase in progesterone secretion by the corpus luteum ✓**
- D. Rise in maternal serum estrogen levels

Q287. Which statements about placental steroid synthesis is most accurate?

- A. The placenta lacks 17 α -hydroxylase and cannot synthesize estrogens directly from cholesterol without fetal adrenal precursors. ✓**
- B. The placenta synthesizes progesterone entirely from fetal cholesterol supplied during the third trimester.
- C. Estriol production relies exclusively on placental enzymes without requiring fetal involvement.

D. Placental steroidogenesis is independent of maternal cholesterol supply.

Q288. Which hormones is primarily responsible for growth of lobuloalveolar tissue of mammary glands during pregnancy?

- A. Estrogen
- B. Progesterone ✓**
- C. Oxytocin
- D. Prolactin

Q289. Which cells in the testis are directly responsible for spermatogenesis?

- A. Leydig cells
- B. Sertoli cells
- C. Spermatogonia ✓**
- D. Myoid cells

Q290. Which hormones is most critical for the initiation and maintenance of spermatogenesis?

- A. Testosterone and Follicle-Stimulating Hormone (FSH) ✓**
- B. Oxytocin and Prolactin
- C. Aldosterone and Cortisol
- D. Insulin-like Growth Factor 1 and Thyroxine

Q291. Which testicular cell types is primarily stimulated by FSH to support spermatogenesis?

- A. Leydig cells
- B. Sertoli cells ✓**
- C. Spermatogonia
- D. Myoid cells

Q292. Which hormones stimulates Leydig cells to produce testosterone?

- A. Follicle-stimulating hormone (FSH)
- B. Luteinizing hormone (LH) ✓**
- C. Adrenocorticotrophic hormone (ACTH)
- D. Growth hormone (GH)

Q293. Which is the major site of conversion of testosterone to dihydrotestosterone (DHT) in males?

- A. Testes
- B. Prostate gland ✓**

- C. Anterior pituitary
- D. Adrenal cortex

Q294. Which is the primary inhibitory regulator of FSH secretion in males?

- A. Inhibin B ✓**
- B. Testosterone
- C. Estradiol
- D. GnRH

Q295. Which represents the correct sequence of cell types in spermatogenesis?

- A. Spermatogonium -> Primary spermatocyte -> Secondary spermatocyte -> Spermatid -> Spermatozoon ✓**
- B. Primary spermatocyte -> Spermatogonium -> Secondary spermatocyte -> Spermatozoon -> Spermatid
- C. Spermatogonium -> Secondary spermatocyte -> Primary spermatocyte -> Spermatid -> Spermatozoon
- D. Spermatid -> Spermatogonium -> Primary spermatocyte -> Secondary spermatocyte -> Spermatozoon

Q296. Which best describes the function of the blood–testis barrier formed by Sertoli cells?

- A. Protects developing sperm from autoimmune reactions by isolating them from the systemic immune system ✓**
- B. Directly produces and secretes testosterone into the seminiferous tubules
- C. Facilitates the process of spermiation by releasing mature spermatozoa into the lumen
- D. Provides nutritional support solely through the active transport of glucose and electrolytes

Q297. Which statements correctly describes the negative feedback loop controlling testosterone levels?

- A. Testosterone directly inhibits the secretion of both GnRH from the hypothalamus and LH from the anterior pituitary. ✓**
- B. Testosterone inhibits the anterior pituitary directly, but stimulates the hypothalamus to release more GnRH.
- C. Inhibin, released by the testes, acts on the hypothalamus to decrease testosterone synthesis.
- D. LH directly inhibits the testes through a short-loop negative feedback mechanism.

Q298. Which responses occurs during the cephalic phase of digestion as a result of vagus nerve stimulation?

- A. Increased gastric acid secretion ✓**
- B. Decreased pancreatic enzyme release

- C. Inhibition of salivation
- D. Relaxation of the pyloric sphincter

Q299. Which mechanisms is most likely responsible for impaired gastric emptying of liquids in a patient after vagotomy?

- A. Loss of receptive relaxation and accommodation of the proximal stomach ✓**
- B. Pyloric spasm due to unopposed sympathetic stimulation
- C. Decreased secretion of motilin by duodenal mucosa
- D. Destruction of the interstitial cells of Cajal in the gastric corpus

Q300. Which statements regarding saliva is true under resting conditions?

- A. Saliva is hypotonic relative to plasma due to active reabsorption of Na⁺ and Cl⁻ exceeding water reabsorption in the salivary ducts. ✓**
- B. Saliva is isotonic relative to plasma because the ductal epithelium is highly permeable to water.
- C. Saliva secretion is primarily regulated by the enteric nervous system without autonomic input.
- D. Saliva contains a high concentration of active enzymes that begin the digestion of proteins.

Q301. Which best describes the “Law of the intestine”?

- A. Peristalsis involves relaxation ascendingly and contraction descendingly to propel contents
- B. Peristalsis involves contraction above a bolus and relaxation below it, propelling contents aborally ✓**
- C. Segmentation contractions occur exclusively in the proximal small intestine to mix chyme
- D. Mass movements occur continuously throughout the small and large intestines to clear waste

Q302. Which best describes innate immunity?

- A. It is non-specific and provides immediate defense against infection upon first exposure. ✓**
- B. It relies on T and B lymphocytes to generate long-lasting immunological memory.
- C. It is highly specific to a particular pathogen and improves with repeated exposures.
- D. It is acquired passively through the transfer of maternal antibodies.

Q303. Which features primarily distinguishes adaptive immunity from innate immunity?

- A. Presence of immunologic memory and high specificity ✓**
- B. Response time within minutes of pathogen exposure
- C. Utilization of physical barriers like the skin and mucous membranes
- D. Lack of improvement in response magnitude upon repeated exposure

Q304. Which cells are primarily responsible for cell-mediated immunity?

- A. B lymphocytes
- B. T lymphocytes ✓**
- C. Erythrocytes
- D. Neutrophils

Q305. Which characteristics mainly distinguishes the primary immune response from the secondary immune response?

- A. A shorter lag phase and higher antibody titer due to the presence of memory cells ✓**
- B. A slower rate of antibody production due to the initial processing by naive T cells
- C. A predominance of IgM antibodies rather than IgG antibodies throughout the response
- D. A lower affinity of antibodies due to somatic hypermutation occurring only in the primary response

Q306. Which is not a component of the innate immune system?

- A. T lymphocytes ✓**
- B. Macrophage
- C. Neutrophil
- D. Complement proteins

Q307. Which pairs is correctly matched regarding immunity type?

- A. Natural passive immunity: Antibodies passed from mother to fetus via placenta ✓**
- B. Artificial active immunity: Injection of preformed antibodies for immediate protection
- C. Natural active immunity: Vaccination with a weakened pathogen
- D. Artificial passive immunity: Recovery from a natural infection

Q308. Which is the first antibody produced in the primary immune response?

- A. IgM ✓**
- B. IgG
- C. IgA
- D. IgE

Q309. Which is not a mechanism by which the complement system enhances immune defense?

- A. Opsonization of pathogens to facilitate phagocytosis
- B. Direct lysis of target cells via the Membrane Attack Complex
- C. Induction of antibody class switching in mature B cells ✓**
- D. Recruitment and activation of inflammatory cells via anaphylatoxins

Q310. Which type of immunity is conferred to a newborn baby through maternal antibodies transferred across the placenta?

- A. Active natural immunity
- B. Passive natural immunity ✓**
- C. Active artificial immunity
- D. Passive artificial immunity

Q311. Which immunological abnormalities is most likely to be observed in an infant born with a congenital defect of the thymus?

- A. Deficient cell-mediated immunity with normal B-cell numbers ✓**
- B. Deficient humoral immunity with absent B cells
- C. Combined immunodeficiency with absent T and B cells
- D. Normal T-cell function with defective phagocytosis

Q312. Which functions explains why antigen-presenting cells (APCs) are crucial in the immune response?

- A. Processing and presenting endogenous antigens via MHC class I molecules to CD4+ helper T cells
- B. Processing and presenting exogenous antigens via MHC class II molecules to CD4+ helper T cells ✓**
- C. Directly producing pathogen-specific antibodies without T cell assistance
- D. Phagocytosing pathogens and destroying them exclusively through non-specific complement activation

Q313. Which type of immune mechanism primarily mediates serum sickness that develops after administration of antiserum?

- A. Type I hypersensitivity reaction involving IgE-mediated mast cell degranulation
- B. Type II hypersensitivity reaction involving antibody-dependent cellular cytotoxicity
- C. Type III hypersensitivity reaction involving circulating immune complex deposition ✓**
- D. Type IV hypersensitivity reaction involving T cell-mediated delayed-type hypersensitivity

Q314. Which part of the brain acts as the principal center for thermoregulation in humans?

- A. Medulla oblongata
- B. Hypothalamus ✓**
- C. Thalamus
- D. Cerebellum

Q315. Which responses is most likely to occur when core body temperature decreases?

A. Cutaneous vasoconstriction to minimize heat loss ✓

- B. Skeletal muscle relaxation to conserve energy
- C. Inhibition of thyroid hormone release to lower metabolic rate
- D. Vasodilation of blood vessels in the extremities to maintain core temperature

Q316. Which mechanism is primarily responsible for heat loss at rest under normal environmental conditions?

A. Radiation ✓

- B. Evaporation
- C. Conduction
- D. Convection

Q317. Which physiological effects occurs when the anterior hypothalamus is stimulated?

A. Sweating and cutaneous vasodilation ✓

- B. Shivering and cutaneous vasoconstriction
- C. Decreased heart rate and decreased blood pressure
- D. Increased appetite and weight gain

Q318. Which physiological mechanisms helps prevent an excessive rise in body temperature during exposure to high environmental temperatures?

A. Increased cutaneous vasodilation leading to enhanced radiant and convective heat loss ✓

- B. Constriction of blood vessels in the skin to trap heat in the core
- C. Shivering thermogenesis to dissipate internal thermal energy
- D. Decreased sweat production to conserve core body water

Q319. Which describes a behavioral mechanism of thermoregulation?

- A. Shivering to generate internal heat
- B. Sweating to promote evaporative cooling
- C. Putting on a heavy sweater when feeling cold ✓**
- D. Vasoconstriction of cutaneous blood vessels

Q320. Which hormonal response enhances long-term heat production during cold exposure?

A. Increased secretion of thyroid hormones (T3 and T4) ✓

- B. Immediate surge of parathyroid hormone
- C. Decreased secretion of insulin
- D. Suppression of antidiuretic hormone

Q321. Which is a local cutaneous thermal reflex?

- A. Axon reflex vasodilation ✓**
- B. Shivering thermogenesis
- C. Sweating via sympathetic cholinergic activation
- D. Central hypothalamic cooling

Q322. Which mechanism explains why a runner experiences profuse sweating but minimal heat loss in conditions of high humidity?

- A. High ambient humidity decreases the vapor pressure gradient between the skin and the air, impairing sweat evaporation. ✓**
- B. High humidity causes cutaneous vasoconstriction, which traps metabolic heat within the core.
- C. High humidity stimulates sympathetic cholinergic fibers to suppress eccrine sweat gland secretion.
- D. High humidity increases the convective heat transfer coefficient, warming the body faster than sweat can cool it.

Q323. Which correctly describes heat stroke?

- A. A severe form of hyperthermia characterized by a core body temperature above 40°C (104°F) accompanied by central nervous system dysfunction ✓**
- B. A condition caused by excessive salt depletion without significant water loss during heavy exercise
- C. A mild elevation in body temperature easily managed by oral rehydration and passive cooling alone
- D. A primary failure of the hypothalamus that results in severe hypothermia due to cold exposure

Q324. Which mechanisms is most important in preventing a dangerous fall in core temperature during prolonged cold exposure?

- A. Shivering thermogenesis and cutaneous vasoconstriction ✓**
- B. Decreased metabolic rate and peripheral vasodilation
- C. Inhibition of thyroid hormone release and sweating
- D. Skeletal muscle relaxation and piloerection

Q325. During fever, Which events occurs first following pyrogen exposure?

- A. Release of endogenous pyrogens such as IL-1 and IL-6 into the circulation ✓**
- B. Increased production of prostaglandins E₂ (PGE₂) in the preoptic area of the anterior hypothalamus
- C. Vasoconstriction and behavioral changes to increase heat conservation and production
- D. Elevation of the hypothalamic set-point temperature

Q326. Which part of the brain is primarily responsible for maintenance of posture and equilibrium?

- A. Cerebrum
- B. Cerebellum ✓**
- C. Hypothalamus
- D. Medulla oblongata

Q327. Which function of the basal ganglia plays an important role in motor control?

- A. Selection and initiation of voluntary movement ✓**
- B. Direct relay of sensory information to the cerebral cortex
- C. Maintenance of posture and balance via vestibulospinal reflexes
- D. Execution of rapid ballistic movements without feedback

Q328. Which function is primarily controlled by the vestibulocerebellum (flocculonodular lobe)?

- A. Balance and equilibrium ✓**
- B. Planning and initiation of voluntary movement
- C. Fine motor control of distal limbs
- D. Velocity and trajectory of ongoing limb movements

Q329. Which structures is most important for maintaining muscle tone?

- A. Muscle spindles ✓**
- B. Golgi tendon organs
- C. Pacinian corpuscles
- D. Free nerve endings

Q330. Which characteristic features occurs as a result of cerebellar damage?

- A. Intention tremor ✓**
- B. Resting tremor
- C. Spastic paralysis
- D. Flaccid paralysis

Q331. Which pathways is most important in the maintenance of balance by adjusting head and body position relative to gravity?

- A. Vestibulospinal tract ✓**
- B. Corticospinal tract
- C. Rubrospinal tract
- D. Tectospinal tract

Q332. Which concept best describes the maintenance of a stable internal environment in the human body?

- A. Homeostasis ✓**
- B. Hemopoiesis
- C. Hyperemia
- D. Hemostasis

Q333. Which is an example of a negative feedback mechanism?

- A. Oxytocin release intensifying uterine contractions during childbirth
- B. Platelet plug formation during blood clotting
- C. Baroreceptor reflex regulating blood pressure ✓**
- D. The surge of luteinizing hormone (LH) triggered by estrogen before ovulation

Q334. Which system acts as the rapid control system for maintaining internal environment stability?

- A. Nervous system ✓**
- B. Endocrine system
- C. Renal system
- D. Lymphatic system

Q335. Which statements regarding somatic sensations is true?

- A. Meissner corpuscles are rapidly adapting mechanoreceptors involved in detecting low-frequency vibrations and fine touch. ✓**
- B. Free nerve endings are encapsulated receptors primarily responsible for proprioception.
- C. Pacinian corpuscles are slowly adapting receptors that encode the sustained duration of a tactile stimulus.
- D. Type Ia sensory fibers are unmyelinated axons that transmit crude touch and temperature sensations.

Q336. Which sensory receptors is responsible for detecting fine touch and texture?

- A. Pacinian corpuscles
- B. Meissner's corpuscles ✓**
- C. Ruffini endings
- D. Merkel discs

Q337. Which tracts primarily carries pain sensations in the nervous system?

- A. Dorsal columns
- B. Spinothalamic tract ✓**
- C. Corticospinal tract
- D. Tectospinal tract

Q338. Which neurotransmitter is most associated with the transmission of pain in the spinal cord?

- A. Acetylcholine
- B. Substance P ✓**
- C. GABA
- D. Dopamine

Q339. Which type of nerve fibers mainly mediate pain caused by tissue injury and inflammation?

- A. A-beta fibers
- B. A-delta fibers
- C. C fibers ✓**
- D. Ia fibers

Q340. Which is true about headache caused by tension or muscle contraction?

- A. It is typically bilateral, steady, and non-pulsatile. ✓**
- B. It is usually unilateral and throbbing in nature.
- C. It is commonly accompanied by severe nausea and vomiting.
- D. It is exacerbated by routine physical activity such as walking.

Q341. Which stimulus primarily increases ventilation at high altitude?

- A. Arterial hypoxemia stimulating peripheral chemoreceptors ✓**
- B. Metabolic alkalosis stimulating central chemoreceptors
- C. Hypercapnia resulting from hyperventilation
- D. Decreased core body temperature stimulating thermoreceptors

Q342. Which acute physiological changes occurs during ascent to high altitude?

- A. Increased heart rate and hyperventilation ✓**
- B. Decreased cardiac output and metabolic alkalosis
- C. Increased arterial oxygen saturation and bradycardia
- D. Decreased pulmonary artery pressure and polycythemia

Q343. Which adaptations occurs in response to chronic exposure to high altitude?

- A. Increased erythropoietin production leading to polycythemia ✓**
- B. Decreased 2,3-DPG levels in red blood cells
- C. Decreased pulmonary ventilation
- D. Metabolic alkalosis due to hyperventilation

Q344. Which best explains why females generally have a lower basal metabolic rate than males?

- A. Females have a higher percentage of adipose tissue, which is less metabolically active than skeletal muscle, compared to males. ✓**
- B. Females have lower core body temperatures, which decrease overall cellular enzyme kinetics.
- C. Females secrete higher levels of thyroxine, which downregulates basal cellular respiration.
- D. Females have a significantly smaller surface-area-to-mass ratio, leading to reduced heat production.

Q345. Which physiological changes occurs during moderate exercise?

- A. Increased systemic vascular resistance
- B. Decreased stroke volume
- C. Increased skeletal muscle blood flow ✓**
- D. Decreased minute ventilation

Q346. Which physiological action is primarily mediated by leptin, a hormone secreted by adipocytes?

- A. Suppression of appetite and increased energy expenditure ✓**
- B. Stimulation of gastric acid secretion
- C. Upregulation of skeletal muscle glucose uptake via insulin-dependent mechanisms
- D. Promotion of adipogenesis and storage of triglycerides in white adipose tissue

Q347. At high altitude, after several days of acclimatization, arterial oxygen content remains subnormal despite increased ventilation. Which best explains this finding?

- A. Hyperventilation causes a significant loss of CO₂, leading to respiratory alkalosis which shifts the oxyhemoglobin dissociation curve to the left, hindering oxygen release to tissues. ✓**
- B. Renal compensation for respiratory alkalosis leads to decreased plasma bicarbonate concentration, which lowers total blood buffering capacity.
- C. Persistent hypobaric hypoxia impairs the bone marrow's ability to synthesize hemoglobin, leading to chronic anemia during the first week.
- D. Hyperventilation increases the work of breathing, consuming a greater fraction of the arterial oxygen supply before it reaches peripheral tissues.

Q348. Which physiological adaptation contributes most to the increase in oxygen delivery to tissues during chronic exposure to high altitude?

- A. Increased erythropoiesis mediated by EPO ✓**
- B. Immediate hyperventilation
- C. Decreased 2,3-BPG in red blood cells
- D. Systemic vasoconstriction

Q349. Which physiological parameters increases with aging rather than decreases?

A. Systolic blood pressure ✓

B. Maximum heart rate

C. Glomerular filtration rate

D. Vital capacity

Q350. Which equations correctly represents the relationship between blood flow, pressure, and resistance?

A. Flow = (Pressure Gradient) / Resistance ✓

B. Flow = Resistance / (Pressure Gradient)

C. Pressure Gradient = Flow $\times$ Resistance²

D. Resistance = Flow $\times$ Pressure Gradient

Duplicate Questions

Question 103 and Question 300 are duplicates:

"Which statements regarding saliva is true under resting conditions?"

Question 147 and Question 302 are duplicates:

"Which best describes innate immunity?"

Question 149 and Question 304 are duplicates:

"Which cells are primarily responsible for cell-mediated immunity?"

Question 152 and Question 306 are duplicates:

"Which is not a component of the innate immune system?"

Question 153 and Question 307 are duplicates:

"Which pairs is correctly matched regarding immunity type?"

Question 154 and Question 308 are duplicates:

"Which is the first antibody produced in the primary immune response?"

Question 159 and Question 314 are duplicates:

"Which part of the brain acts as the principal center for thermoregulation in humans?"

Question 160 and Question 315 are duplicates:

"Which responses is most likely to occur when core body temperature decreases?"

Question 164 and Question 319 are duplicates:

"Which describes a behavioral mechanism of thermoregulation?"

Question 166 and Question 321 are duplicates:

"Which is a local cutaneous thermal reflex?"

Question 168 and Question 323 are duplicates:

"Which correctly describes heat stroke?"

Question 169 and Question 324 are duplicates:

"Which mechanisms is most important in preventing a dangerous fall in core temperature during prolonged cold exposure?"

Question 170 and Question 325 are duplicates:

"During fever, Which events occurs first following pyrogen exposure?"

Question 171 and Question 326 are duplicates:

"Which part of the brain is primarily responsible for maintenance of posture and equilibrium?"

Question 174 and Question 329 are duplicates:

"Which structures is most important for maintaining muscle tone?"

Question 176 and Question 331 are duplicates:

"Which pathways is most important in the maintenance of balance by adjusting head and body position relative to gravity?"

Question 177 and Question 332 are duplicates:

"Which concept best describes the maintenance of a stable internal environment in the human body?"

Question 178 and Question 333 are duplicates:

"Which is an example of a negative feedback mechanism?"

Question 179 and Question 334 are duplicates:

"Which system acts as the rapid control system for maintaining internal environment stability?"

Question 180 and Question 335 are duplicates:

"Which statements regarding somatic sensations is true?"

Question 181 and Question 336 are duplicates:

"Which sensory receptors is responsible for detecting fine touch and texture?"

Question 183 and Question 338 are duplicates:

"Which neurotransmitter is most associated with the transmission of pain in the spinal cord?"

Question 185 and Question 340 are duplicates:

"Which is true about headache caused by tension or muscle contraction?"

Question 187 and Question 342 are duplicates:

"Which acute physiological changes occurs during ascent to high altitude?"

Question 189 and Question 344 are duplicates:

"Which best explains why females generally have a lower basal metabolic rate than males?"

Question 192 and Question 347 are duplicates:

"At high altitude, after several days of acclimatization, arterial oxygen content remains subnormal despite increased ventilation. Which best explains this finding?"

Question 195 and Question 350 are duplicates:

"Which equations correctly represents the relationship between blood flow, pressure, and resistance?"